

The background of the entire page is a monochromatic red. It features a profile of a woman's head facing left. Overlaid on the back of her head is a complex, circular, futuristic interface. In the center of this interface is a large, stylized 'AI' logo. The interface has various concentric circles, lines, and smaller circular elements, giving it a high-tech, artificial intelligence feel. The overall aesthetic is futuristic and technological.

nasscom

Compendium of India's Top AI Research

Proceedings from the Research Track at the
Nasscom Agentic AI Confluence 2025

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Foreword

Artificial Intelligence (AI) has emerged as one of the most transformative forces of our time, reshaping industries, redefining possibilities, and reimagining the way we live and work. India, with its dynamic ecosystem of corporates, startups, and academic institutions, is at the forefront of this revolution. The Nasscom AI Research Compendium is a celebration of this collective ingenuity, showcasing the nation's most promising research assets and highlighting the strides we are making in advancing AI for both business and society.

This compendium brings together 16 top research assets, carefully shortlisted from an impressive 150+ nominations received from across the country. These nominations came from corporates pushing the boundaries of applied AI, startups driving innovation with agility, and educational institutes nurturing the next generation of talent. The final selection represents the very best of India's AI ecosystem—projects that embody technical excellence, creativity, and real-world societal impact.

The showcased assets were unveiled at the Nasscom Agentic AI Confluence, held on 7th October 2025 in Bangalore, India. This event served as a platform to spotlight the depth and diversity of AI research being undertaken across the nation. From breakthroughs in medical science to innovations in agriculture, from advancements in autonomous vehicles to pioneering work in Green IT, and from specially tuned AI foundation models to domain-specific applications, these assets reflect the breadth of possibilities that AI unlocks. Each project demonstrates how Indian researchers and innovators are addressing pressing challenges while also opening new avenues for growth and opportunity.

Highlights from this compendium include medical science research that enhances intraoperative detection of cancer resection margins and affordable diagnostics for vertigo, promising improved patient outcomes. In agriculture, AI-driven solutions are empowering farmers to optimize yields,

manage resources more efficiently, and build resilience against climate change. Autonomous vehicle research showcases India's growing capabilities in mobility innovation, while Green AI initiatives emphasize sustainability in the digital age through distilled large language models optimized for compute.

Equally noteworthy are the specially tuned AI models, such as Tiny Time Mixers (TTM) and Geo-Spatial Foundation Models, which demonstrate how nuanced, domain-specific approaches can deliver superior performance. Other innovations featured include transforming literature into visual multilingual & accessible formats, information-theoretic approaches to scaling speech AI, expediting legal processes through document automation, generative models for data visualization and dimensionality reduction, autonomous incident management, AI agent orchestration and also warehouse automation. Several solutions also focus on making AI cost-competitive and easier to consume, such as paper-based analytical devices for molecular sensing and automated government form-filling for aged and monolingual citizens.

This compendium offers an accessible window into the future not just for high tech consumers but for humanity at large. It is not only a collection of technical achievements but also a narrative of how AI is becoming

deeply embedded in our everyday lives. By presenting these research assets in a clear and engaging manner, the compendium seeks to inspire curiosity, foster dialogue, and encourage broader participation in the AI journey.

The significance of this compendium lies not just in the research it showcases, but in the collaborative spirit it represents. AI innovation in India is being driven by a unique synergy between academia, industry, and startups. This collaboration ensures that research is not confined to laboratories but is translated into solutions that address real-world challenges. It is this spirit of partnership and shared vision that will continue to propel India forward as a global leader in AI.

As we look ahead, the journey of AI in India will be defined by inclusivity, sustainability, and innovation. The Nasscom AI Research Compendium is a milestone in this journey—a testament to the talent, creativity, and determination of our researchers and innovators. May it serve as a source of inspiration for all readers, and as a reminder of the immense potential that lies within India's AI ecosystem.

Ankit Bose

Head, Nasscom AI

Preface

The Nasscom Agentic AI Confluence 2025 is where India's AI visionaries, innovators, and policymakers converge to shape the age of autonomous intelligence. This one-day event was organized with an intent to deliver a strategic blueprint for building robust, scalable, and trustworthy AI systems. From cutting-edge architectures to governance frameworks, for businesses to move beyond pilots and architect the autonomous future.

As Nasscom, India's premier industry body for technology, with a vision to create, nurture and foster Research ecosystem across academia, start ups and corporates, we embarked on the mission to host Research Track at this years Agentic AI Confluence event. This Research showcase was a unique opportunity to witness how India's researchers are pushing the boundaries of science and technology—turning theoretical breakthroughs into tangible solutions that can transform industries and society. Towards this, we invited Research assets across industry verticals which received overwhelming response with over 150 top

class assets submissions. Through careful evaluation of these assets, considering their innovation quotient, social impact, practical realizability, TRL level and overall ROI, 16 of these are screened in for it to be presented at this conference both as Research Showcase Poster as well as presentation to invited elite audience. Our goal is to showcase these impactful works to a curated network of industry leaders, investment partners, and stakeholders. This was a platform to showcase and propel India's most innovative AI research from the lab into the market and encourage cross pollination, collaboration and joint development opportunities.

The AI Research Showcase at the Nasscom Agentic AI Confluence was very successful event with all 16 researchers across academia, industry and start-ups, travelling from across India, to present their innovation. This showcase did spotlight cutting-edge research in Agentic AI, spanning diverse industry verticals such as healthcare, retail, transportation, agriculture, and foundational models. Each

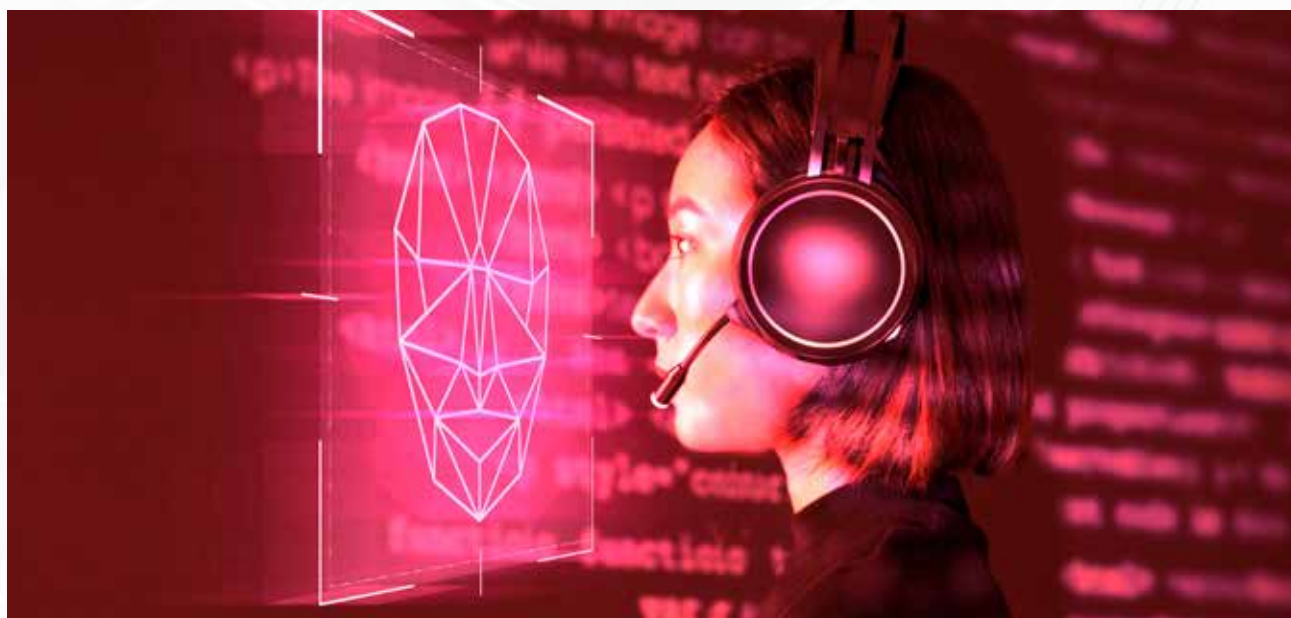
presentation highlighted novel approaches with the potential for significant social and economic impact, with most projects

already at TRL 5, demonstrating readiness beyond concept and moving toward real-world productization.

Expected outcomes from this Research Asset & innovation showcase include:



This is more than just a showcase; it's a platform designed to foster a direct connection between your groundbreaking research and the market needs of today.



Global Poetics AI

Making Literature Visual, Multilingual & Accessible



Presenter

Dr. Sriparna Saha, Associate Professor,
Department of Computer Science &
Engineering Indian Institute of Technology
Patna



Collaborators

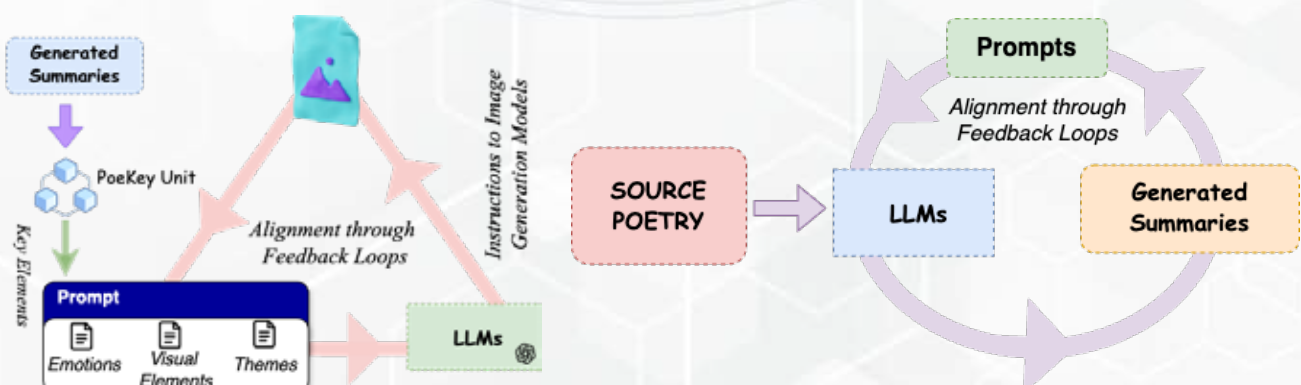
This research is in collaboration with
Koustava Goswami (Adobe Research),
Joseph K J (Adobe Research) & Dr. Sofia
Jamil IIT Patna



Summary

Dr. Saha presented on accessibility, understanding, and cultural preservation depicted in poetry through an AI storytelling tool that bridges literature, art, and technology. The USP here is encoding Indian poetic traditions of 21 languages into modern AI storytelling, ensuring global reach without cultural loss by generating culturally grounded visuals from Indian poems. Moreover, she works on generating images for English poems as well to enhance comprehension. This research asset can offer integrations for digital publishing, interactive e-books, and cultural heritage apps with interactive storytelling and visualization of literature for classrooms.

Prompt Alignment via Human Feedback





Methodology

Text-to-image generation has shown great promise in recent years; however, when applied to poetry, it faces unique challenges due to the abstract, layered, and metaphorical nature of poetic language. To address this, we present a series of frameworks and techniques that advance the task of poem-to-image generation. First, this research introduces PoemToPixel, which incorporates prompt tuning and the PoeKey algorithm to extract emotions, visual elements, and themes from poems for diffusion-based image generation. Alongside this, it releases MiniPo, a multimodal dataset of 1,001 children's poems and images, supporting diverse evaluation. Building on this foundation, it proposes PoemTale Diffusion, a training-free approach that segments poems based on entities and emotions. It integrates a multi-stage prompt refinement loop for LLMs to generate poem descriptions. It additionally employs consistent self-attention in diffusion models to produce semantically coherent images. To facilitate further research in this area, the work contributes the P4I (PoemForImage) dataset with 1,111 curated poems and expert annotations. Finally, it introduces Weighted Prompt Manipulation (WPM), a novel zero-shot technique that modifies attention weights and embeddings within diffusion models, enabling dynamic control over word importance and allowing users to refine generated images to better capture

poetic nuance. Together, these contributions establish new methodologies, datasets, and tools for bridging the gap between poetic expression and visual imagination, paving the way for richer multimodal creativity in the literary domain. Extending this vision to cultural contexts, the research turns to Indian poetry, which is known for its linguistic complexity and deep cultural resonance, spanning thousands of years of heritage. Its layered meanings, cultural allusions, and sophisticated grammatical constructions often pose challenges for comprehension, particularly for non-native speakers or readers unfamiliar with its context and language. Despite this cultural richness, existing works on poetry have largely overlooked Indian language poems. To address this, the research proposes the Translation and Image Generation (TAI) framework, which leverages Large Language Models (LLMs) and Latent Diffusion Models through appropriate prompt tuning.



Novelty

Encodes Indian poetic traditions into 21 languages and integrates aesthetics into modern AI storytelling, ensuring global reach without cultural loss. This research moves beyond academia by offering scalable APIs, SaaS tools, and B2B licensing tailored for publishing and education. It enhances the accessibility of poetry across languages, makes literature visually engaging, and helps preserve cultural heritage while appealing to global audiences.



Validation

Results

	Images without Summarisation	Images without Prompt Tuning	Images with PoemToPixel Approach		Images without Summarisation	Images without Prompt Tuning	Images with PoemToPixel Approach	
1) Suburban Sonnet by Gwen Harwood								1) Five Little Ducks
2) Aunt Helen by T.S. Eliot								2) POLLY PUT THE KETTLE ON
3) The Spanish Needle by Claude McKay								3) Pussy cat, Pussy cat
Theme	✗	✓	✓		✗	✓	✓	Theme
Elements	✗	✗	✓		✓	✓	✓	Elements
Meaning	✗	✗	✓		✗	✗	✓	Meaning
		a)			b)			



Impact

This comprehensive experimental evaluation, including both human and quantitative assessments, demonstrates the superiority of image generation frameworks, outperforming strong baselines. To further address the scarcity of resources for Indian-language poetry, the research introduces the Morphologically Rich Indian Language Poems (MorphoVerse) dataset, comprising 1,570 poems across 21 low-resource Indian languages.



Future Scope

Current creative AI tools focus on monolingual (mostly English) or Western-centric poetry and art generation. This asset is the first to bridge multilingual poetry (English + Indian vernaculars) with visual storytelling and can be enhanced to support more Indian languages. As this research integrates with NLP-powered creative tools, storytelling platforms, digital classrooms, poetry/arts education, and multilingual content creation systems, it has future scope in EdTech, publishing, digital learning platforms, cultural organizations, and entertainment sectors that seek creative, interactive, and culturally grounded AI tools.

Clinical Device for Real Time Intraoperative Detection of Cancer Resection Margin with AI



Presenter

Dr. Subhashish Sarkar MBBS, MS (General Surgery), Assistant Professor Department of General Surgery College of Medicine and Sagore Dutta Hospital, Kolkata, West Bengal



Collaborators

Dr. Ritabrata Saha, Avishek Dutta, College of Medicine & Sagore Dutta Hospital, Kolkata. Prof. Ranjan Ganguly, Prof. Apuba K Santra Jadavpur University, Kolkata and Prof. Utpal Garain, Indian Statistical Institute, Kolkata.



Summary

Cancer carries not only a clinical burden but also profound social and psychological stigma. Surgery is one of the most physically and emotionally demanding parts of the patient journey. This device reduces the risk of leaving behind residual cancerous tissue

that decreases post-operative anxiety and stress for patients, provides surgeons with confidence and precision in resections, and improves long-term survival and quality of life. By reducing recurrence rates, the device contributes to better public health outcomes and lower healthcare system burden, aligning with global health equity goals.



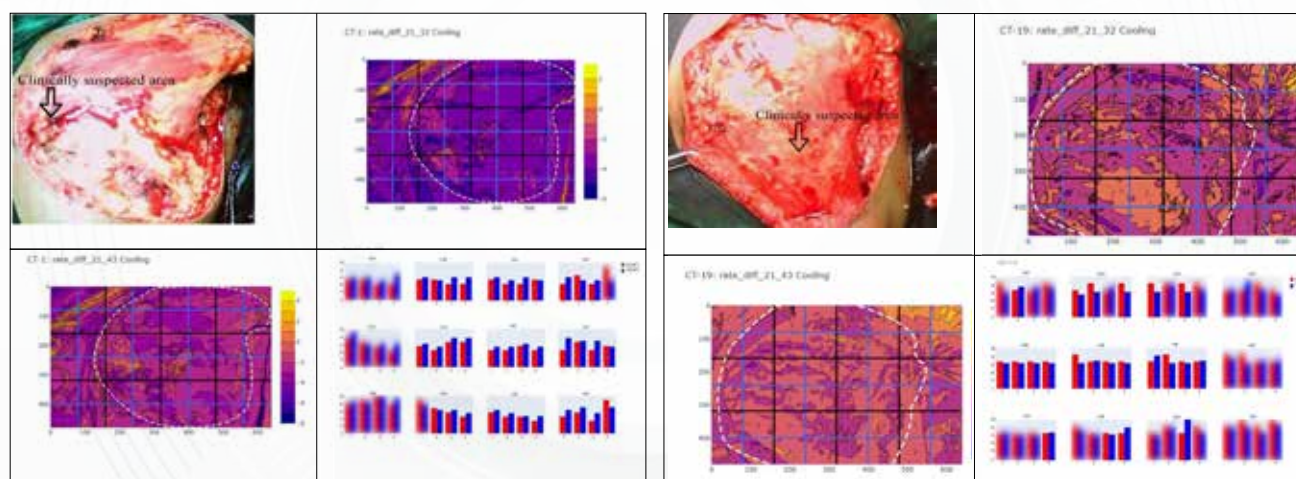
Methodology

Challenges in current methods include imprecise visual and tactile cues, or slow, invasive imaging methods (e.g. frozen section analysis, intraoperative MRI, contrast-based NIR imaging) which fail to detect residual tumor tissue during surgery and causes margin-positive resections (R1) resulting in higher rates of cancer recurrence and increased patient morbidity and mortality and results in substantial post-operative costs for chemo-radiation therapies. By offering non-invasive, real-time, and accurate detection of residual tumor tissue, the proposed device improves surgical success rates and reduces downstream treatment costs, shortens hospital stays and accelerates recovery.

Core Algorithm & Methodology includes proprietary AI-powered thermal image processing engine - detects microscopic residual malignancies with high sensitivity. First it captures thermal response, following brief IR heating, thermal video of the cooling phase is recorded at high frame rates. In the next phase which is feature extraction, advanced computer vision pipelines extract spatiotemporal features (cooling rate gradients and pixel-wise decay constants from each tissue zone)

followed by AI classification - Trained machine learning model mapping these features to probabilistic malignancy scores for each region of interest. With real-time visualization by overlaying color-coded risk maps on live surgical visuals, it enables immediate decision-making by the surgeon. To ensure both clinical usability and high diagnostic performance, the platform comprises of FLIR thermal imaging camera (FLIR A700 24°), infrared heating modules (1 kW), custom high-performance workstation and clinical safety compliance layer.

Observation



Novelty

This research innovative solution includes active thermal imaging platform purpose-built for real-time intraoperative detection of residual cancer tissue with Infrared thermal excitation, high-resolution thermal and visual imaging and AI-driven image analysis pipeline. This asset is the first real-time intraoperative cancer margin detection system that integrates seamlessly into existing surgical workflows.

It addresses a critical gap in oncologic surgery by combining Clinical Precision which significantly reduces false negatives, lowering the incidence of margin-positive (R1) resections and improving long-term survival outcomes, Operational Efficiency which provides margin status within ~90 seconds, eliminating hours-long delays associated with frozen sections or MRI scans, Economic Advantage by avoiding the need for repeat surgeries and costly post-operative chemo/radiation therapies, resulting in substantial savings for hospitals, payers, and patients and Ease of Use

through non-invasive, surgeon-friendly, and compatible with standard OT infrastructure, ensuring effortless adoption.

By merging clinical excellence with operational and financial efficiency, the system creates a triple-value proposition for hospitals, clinicians, and patients alike.

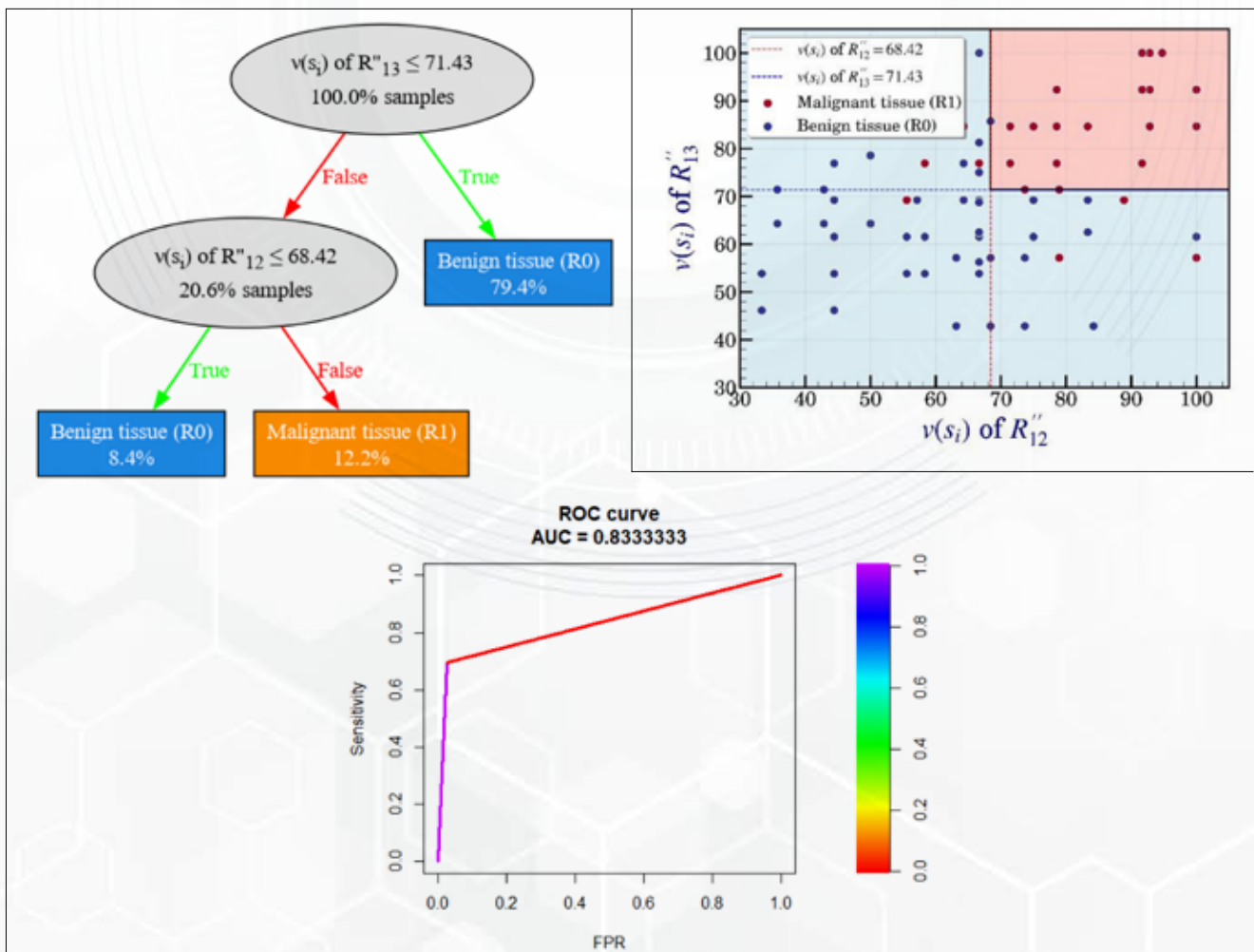


Validation

The device has already been successfully deployed during live breast cancer surgeries at Sagore Dutta Medical College to detect

residual tumor tissue in the chest bed. The entire procedure from thermal excitation to processed margin visualization takes ~90 seconds, allowing uninterrupted surgical flow. Surgeons receive direct intraoperative feedback to guide precision resections in real time, drastically reducing the risk of leaving malignant tissue behind. This positions the device as a decision-support companion for oncologic surgeons, rather than a disruptive add-on. This proven intraoperative integration demonstrates clinical readiness and operational scalability, paving the way for commercialization and adoption in tertiary care centers globally.

Statistical Analysis





Impact

The global intraoperative imaging market is projected to reach \$5.4–\$5.65 billion by 2030, driven by the rising burden of cancer worldwide and the growing demand for precision surgery solutions. From market opportunity perspective over 17 million cancer surgeries annually require intraoperative margin assessment, with volumes increasing alongside global cancer incidence and hence hospitals are actively seeking real-time, non-invasive, and workflow-compatible intraoperative imaging technologies to reduce surgical errors and improve patient outcomes, however existing CT, MRI, and frozen section methods are slow, invasive, and costly, creating a clear adoption runway for rapid, AI-powered margin detection systems.



Future Scope

First and foremost, this proven AI-based architecture allows extension to multiple solid cancer types (breast, oral, head & neck, gastrointestinal, gynaecological, thoracic and neurosurgery) and the modular system design supports customization for various surgical specialties without major hardware redesign.

Future research scope also includes Technology Maturation with complete integration of FLIR thermal imaging software with custom AI algorithms and frontend visualization tools Enhance user interface for real-time surgeon interaction and margin mapping, followed by Clinical Validation to conduct large-scale clinical studies across multiple hospitals and patient cohorts and achieve multi-site validation to demonstrate generalizability and reproducibility. This research also needs to have Regulatory Readiness through regulatory pathway preparation (DCGI/CDSCO) taking care of safety, performance, clinical efficacy, risk management, and ISO 13485-compliant quality systems.



Vestibulus Frenzel Device

Affordable Diagnostics for Vertigo



Presenter

Mr. Adarsh Pandey, PhD Scholar, from IIIT Delhi, presented Vestibulus, a multimodal AI-powered diagnostic for vertigo



Collaborators

This research is in collaboration with Dr Rajiv Ratn Shah and Dr Jainendra Shukla from IIIT Delhi.



Summary

Vertigo is a symptom associated with more than 40 inner-ear and neurological disorders. It is a highly prevalent and debilitating condition, affecting more than one in every 200 individuals. In India alone, nearly 180 million people suffer from balance and dizziness disorders. Among these, Benign Paroxysmal Positional Vertigo (BPPV) is the most common, yet remains frequently misdiagnosed. Such misdiagnosis leads to prolonged disability, increased

fall risk, anxiety, and unnecessary medical investigations.

A key barrier is the absence of affordable and accessible diagnostic tools, particularly in low-resource healthcare environments. Vestibulus, the presented research asset, is a low-cost, portable, and AI-enabled Video Frenzel device designed to diagnose vestibular disorders such as BPPV, vestibular neuritis, and central vertigo.

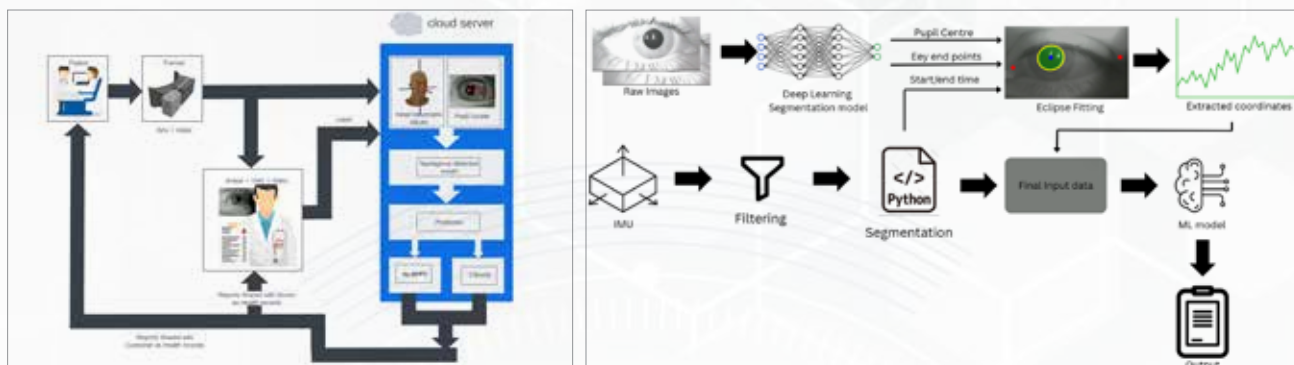


Methodology

Current vertigo assessments are largely manual and rely on inducing vertigo attacks, followed by subjective visual evaluation of eye movements. The proposed solution addresses this gap by integrating infrared video-based eye tracking with motion sensing to objectively capture nystagmus and head movements.

AI models embedded within the device automate analysis and assist clinicians by providing rapid, reliable, and interpretable diagnostic insights. The system is designed to function in both clinical and remote settings, enabling specialist-grade evaluation beyond major hospitals.

Fig. 2 Complete Solution Overview



Novelty

The core innovation lies in combining **affordability, portability, and AI-driven automation** within a single diagnostic platform, priced at nearly one-fifth of the global alternatives. Unlike existing bulky and expensive Frenzel systems, Vestibulus is lightweight, mobile-connected, and operable even by trained assistants, making it suitable for rural, resource-constrained environments.

A key technical advance is the **fusion of eye and head movement features**, which significantly enhances diagnostic accuracy. This is the first affordable Frenzel device that integrates multimodal sensing with agentic AI capable of classification, maneuver suggestions, and remote clinical assistance.

The solution directly aligns with SDG 3 (Good Health & Well-Being), SDG 9 (Industry, Innovation & Infrastructure), and SDG 10 (Reduced Inequalities).



Validation

Currently at **TRL 6**, the prototype has been tested in real-world clinical environments,

with neurologists actively using it for patient diagnosis and treatment. Clinicians report strong usability, portability, and valuable decision-support capability.

The device is undergoing formal clinical trials and is backed by a provisional patent. Plans for full IP protection and regulatory approvals, including CDSCO Class A and CE marking, are in progress.

The evaluation framework includes:

01

UNet (ResNet34 encoder) for eye segmentation

02

Classical ML models (LGBM, Random Forest, NuSVC) for classification

03

5-fold GroupKFold cross-validation

04

Metrics: Dice, IoU (segmentation), F1-score, and PR curves (classification)

05

Classical ML models (LGBM, Random Forest, NuSVC) for classification

Metric	Eye Only	Eye+Head
Accuracy	0.71	0.84
F1-Score	0.73	0.85
ROC-AUC	0.63	0.80
Balanced Accuracy	0.70	0.84

CAGR). In India, potential adopters include more than 6,100 ENT specialists and 4,700 neurologists. Despite the high prevalence of vertigo and its severe impact, diagnosis remains challenging due to limited specialist availability (fewer than 200 dedicated experts nationwide) and high device costs (₹2–15 lakh).



Impact

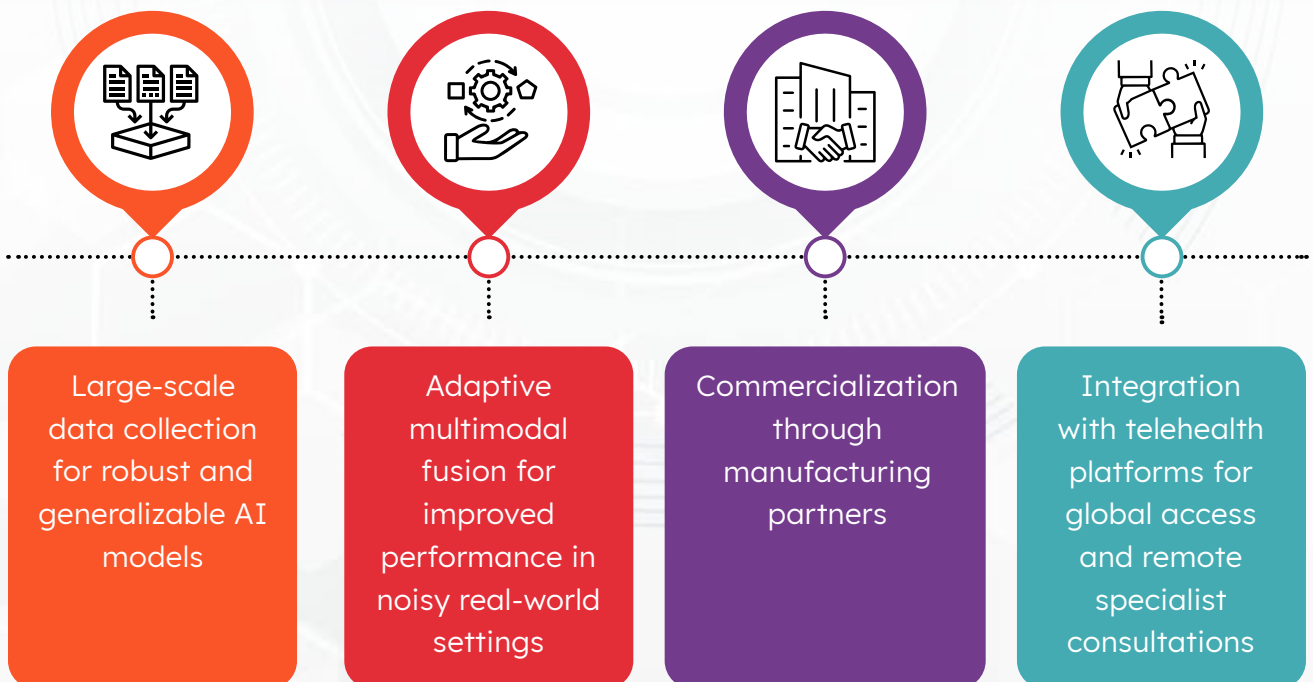
The global vestibular diagnostics market is expected to reach USD 136M by 2028 (5.6%

By offering an accurate, affordable, and portable solution, Vestibulus enables timely diagnosis, minimizes disability, and improves patient outcomes at scale.



Future Scope

Future directions include:



The approach extends beyond BPPV to central vertigo, vestibular neuritis, and vestibular migraine, offering a versatile diagnostic framework for a wide range of vestibular disorders.

Vision-Centric Low-Cost Autonomous Navigation for Ground Vehicles



Presenter

Dr. Santhosh Reddy, Hub Executive Officer / Chief Executive Officer, NMICPS Technology Innovation HUB on Autonomous Navigation Foundation(TiHAN), IIT Hyderabad.



Collaborators

This research is in collaboration with Prof P. Rajalakshmi, Department of EE & AI, Project Director, TiHAN-IIT Hyderabad



Summary

This research presents a step-by-step workflow for Autonomous Vehicle Implementation, pioneering India's first Autonomous Campus Shuttle (ACS), which is operational since August 2023. Driven by AI-powered multi-sensor fusion of LiDAR, GNSS/INS and cameras, the ACS delivers safe, reliable, and eco-friendly transportation. With advanced features

such as obstacle detection, Adaptive Cruise Control, and Autonomous Emergency Braking, the shuttles redefine safety and convenience. Integrated with a mobile app and onboard GUI for real-time tracking.



Methodology

The autonomous driving stack comprises of environment feed with real time feed for traffic condition, weather, pedestrians etc, sensors for Camera, RADAR, LiDAR, GPS, IMU etc, Perception detection for lane, traffic lights, traffic signs, object detection, free space detection and localization, control system comprising of PID (Proportional Integration derivative) and MPC (Model predictive Control) and DWB (Drive by Wire). Perception stack features potholes and speed bump detection along with marked and unmarked cross-walks, in real time, with camera feed and sensors, with LLM trained on campus data set. The Lane detection module detects Lanes, free drive space detection and lane width estimation. Obstacle detection and classification includes Collision warnings – ISO Standard 19237:2017 & Pedestrian Intent prediction. Traffic sign and Traffic light Indian data along with IEEE Dataport

is used for training. YOLO v8 advanced deep learning capabilities provide fast and accurate pothole and speed bump detection, crucial for autonomous vehicles navigating complex Indian roads. The Pedestrian intent system uses a Zed camera for real-time object detection, tracking, and distance estimation with YOLO v8, ByteTrack, and CenterNet. Predicted trajectories inform vehicle decisions, ensuring safety through continuous feedback and adjustment. The autonomous navigation system ensures safe driving by dynamically generating and adjusting paths for overtaking and obstacle avoidance. It pauses to assess obstacles, navigates around them, handles multiple and curved paths, avoids collisions, and smoothly returns to the original route by using PSPNet with ResNet Backbone which enhances driving space segmentation, improving vehicle autonomy and safety in complex urban roads. PSPNet can perform semantic segmentation and ResNet has estimate depth and features. For traffic lights and traffic signs prediction, the data that was used is a combination of data from in and around the campus, IEEE data port and various other resources. Train and fine-tune a deep learning model capable of identifying and categorizing traffic light states (red, green, yellow) with high precision, contributed to improved autonomous driving and urban transportation systems real time. LiDAR processing engine uses advanced mapping, localization, and obstacle detection using real-time point cloud data analysis for environmental understanding. Velodyne sensors scan environment during manual route drives, recording comprehensive 3D spatial data and advanced LiDAR Odometry and Mapping processes recorded data

using ROS .bag files for accuracy. AI based drive by wire interface uses software layer translating high-level navigation commands into precise actuator controls for safe vehicle operation while the communication system uses audio announcement package managing stop notifications, boarding alerts, and comprehensive passenger communication protocols.

Major steps in the workflow includes (a) environmental mapping with LiDAR Data Collection & point cloud generation, (b) Path navigation Integration involves Reference Framework Integration, Real-Time Alignment and continuous localization. (c) Waypoint Collection stage involves dual-run process (the first run generates the baseline map and the second run validates it by confirming waypoints against live LiDAR scans) and visual verification (d) Autonomous navigation includes obstacle detection, collision avoidance, alternate path identification while (e) stops-and-announcement step includes Live Waypoint Monitoring, stop identification and custom instructions.



Novelty

Powered by advanced AI-driven multi-sensor fusion, integrating LiDAR, INS/GNSS, and camera technologies, this campus shuttle is capable of environment perception, mapping, localisation, and autonomous navigation. Its obstacle detection system ensures passenger safety, both in short- and long-range scenarios. The solution primarily consists of Drive-by-

Wire system consisting of Precision steering motor with encoder feedback, electronic throttle control via motor system & linear actuator for reliable braking response along with Perception & Control consisting of 360-degree LiDAR environmental mapping, central PLC coordinating all systems & orion interface managing sensor data flow. With this the ACS can detect & navigate potholes and speedbumps, can detect and classify obstacles and predict pedestrian intent. The custom object detection model uses Custom YOLO v8 while the Pedestrian intent system uses a Zed camera for real-time object detection, tracking, and distance estimation with YOLO v8, ByteTrack, and CenterNet. Automated lane and driving space detection uses PSPNet with ResNet Backbone to enhance driving space segmentation, improving vehicle autonomy and safety in complex urban roads. PSPNet performs semantic segmentation and ResNet has estimate depth and features. ACS also has traffic sign and traffic light detection based on trained YOLO V8 model.



Validation

This Autonomous Campus Shuttle (ACS) service has already transported 10,000+ commuters across 20,000 km, earning 90% user satisfaction. Validated at TRL-9, this initiative demonstrates not just a campus innovation but a transformative step

toward India's smart mobility revolution. It sets new benchmarks in passenger safety, sustainability, and intelligent transport, inspiring the future of autonomous mobility across the nation.



Impact

Integrated with a mobile app and onboard GUI for real-time tracking, and validated at TRL-9, this initiative demonstrates not just a campus innovation but a transformative step toward India's smart mobility revolution. It sets new benchmarks in passenger safety, sustainability, and intelligent transport, inspiring the future of autonomous mobility across the nation.



Future Scope

This deployment establishes a strong foundation for expanding autonomous transportation solutions across other campuses and controlled environments with safe, reliable, and eco-friendly transportation, with advanced features such as obstacle detection, Adaptive Cruise Control, and Autonomous Emergency Braking. This can be extended to cargo vehicles and eventually to passenger vehicles.

LegalTease, GenAI-based Legal Document Automation System



Presenter

Mr. Siddharth Kanungo, Senior Associate IIT Mandi iHub and HCl Foundation Technology Innovation Hub



Collaborators

This research is in collaboration with Mr. Sahil Jaglan, Mr. Naveen Kumar, Ms. Shimona Jaglan and Mr. Somjit Amrit from IIT Mandi iHub and HCl Foundation Technology Innovation Hub.



Summary

This research asset is an AI-powered legal automation system that converts scanned legal documents (paperbooks) into structured, searchable case notes in 2-3 minutes, reducing what traditionally requires 4-6 hours of manual work. A paperbook is a comprehensive compilation of all documents

related to an appeal, including document index, memo of parties, chronology of dates and events, impugned orders, grounds of appeal, annexures, affidavits, and miscellaneous applications. This innovation addresses a critical bottleneck in India's judicial system by automating the tedious manual process of document structuring and case note preparation.



Methodology

The Paperbook AI solution addresses a critical bottleneck in India's judicial system by automating the tedious manual process of converting legal documents into structured case notes. A paperbook is comprehensive compilation of legal documents that forms the foundation of court cases including document index and structure, memo of parties involved, critical dates and timelines, court orders and judgments, grounds of appeal & supporting annexures and affidavits

The solution involves an AI agent first indexing the pages of uploaded documents followed by classifying each section as Petition, Annexure,

FIR, Supporting document etc and then more importantly summarizing each document with exact answers from relevant and appropriate sections across all the documents. Validation

agent then validates the summary generated by previous agent through cross verification and finally case note is generated for the user to edit and finalize.



This innovation delivers significant social benefits across multiple dimensions, such as



Access to Justice: By reducing document processing time from hours to 2-3 minutes, the system enables faster case preparation and legal service delivery, making justice more accessible and affordable for common citizens. Smaller law firms can now compete effectively with larger practices through enhanced efficiency.



Judicial Efficiency: The technology directly tackles India's massive case backlog problem by streamlining workflows that previously consumed substantial professional time. Legal practitioners can redirect their expertise from mundane document processing to higher-value legal analysis and client representation.



Quality Enhancement: The multi-agent verification system reduces human errors in document processing, ensuring more accurate case preparation and better legal outcomes. The collaborative features enable secure information sharing among stakeholders, improving coordination and case quality.



Systemic Transformation: By automating the "tedious bits" of legal work, this solution represents the first step toward comprehensive AI-driven judicial modernization, laying the foundation for broader technological transformation that can ultimately strengthen the rule of law and democratic institutions.



Novelty

Reduced processing time to 2 minutes from days of manual work, five AI agents working in harmony to provide 100% automated conversion from scanned documents to case notes, enhancing existing legal processes without disrupting established practices while maintaining confidentiality with government-grade security protocols and reliability layer to ensure accuracy through systematic validation processes. These five AI agents include Indexing agent for extracting document structure, OCR agent for converting manual writing to text, Summarizing agent to create key insights, Extraction agent to answer legal queries and Verification agent for validating accuracy.

Novelty of this solution includes built-in verification layer for reliability assurance, rapid processing within 2 min, modular scalability with independent agent optimization, Indian legal system expertise with deep understanding of local documents structure and compliance, government grade security with audit trails, encryption & compliance protocols.



Validation

Real-life validation was conducted with actual legal paperbooks from different case types including civil appeals, criminal matters, writ petitions, and special leave petitions, ensuring the system's ability to

handle diverse legal document formats and complexity levels encountered in Indian courts. The multi-agent verification system demonstrates reasonable accuracy in document classification (identifying petition vs. annexure vs. FIR), text extraction from both printed and handwritten sources, and summarization tasks that capture legally relevant information while maintaining source traceability. Legal professionals including advocates, junior lawyers, and legal researchers have participated in validation exercises, comparing automated outputs against manually prepared case notes to establish accuracy benchmarks and identify areas requiring refinement.



Impact

In today's scenario of Legal and judiciary system, legal professionals spend hours manually converting paper books into structured case notes as current tools focus only on judgment analysis, ignoring document workflow automation while India's courts face massive case backlogs due to inefficient preparation processes. From a market opportunities perspective, we have 25 High Courts across India in addition to the Supreme Court system, employing 1.3+ million registered advocates and law firms nationwide. With the latency of pending cases in Indian judiciary system, there is dire need for enhancing existing legal processes without disrupting established practices through workflow automation, maintaining confidentiality with government-grade security protocols.

The innovation in this solution includes Multi Agent AI System (Five specialized collaborative agents), Indexing (To extract document structure), OCR (To convert scans to text), Summarizing (To create key insights), Extraction (To answer legal questions) and Verification (To validate accuracy). This solution approach integrates and enhances existing workflows with secure collaboration, real-time editing and confidentiality-first design.



Future Scope

The future development of Legatease follows a phased approach starting with gradual expansion across the judicial system through pilot deployment in select High Courts, followed by systematic expansion to additional courts based on validated results and user feedback while building adoption among legal professionals. The second phase focuses on integration with existing Court Information Systems

(CIS) and e-courts infrastructure through development of APIs and connectors, enabling seamless data flow and reducing redundant data entry across judicial platforms. The third phase involves expansion to the private sector including private law firms, corporate legal departments, and legal process outsourcing organizations, adapting features for commercial litigation and corporate legal work beyond the public judicial system. Advanced capabilities in the fourth phase include development of multilingual support for regional language documents, advanced analytics for case insights, predictive features to support legal strategy development, and expansion to additional document types beyond paperbooks including contracts, agreements, and legal correspondence. The long-term vision establishes Legatease as a foundational component of India's legal technology ecosystem, contributing to faster case resolution, improved access to justice, and strengthened judicial efficiency across the country while continuously evolving based on user needs and technological advancements.



G-NeuroDAVIS

Generative Model for Data Visualization & Dimensionality Reduction



Presenter

Prof. Dr. Rajat Kumar De, Machine Intelligence Unit, Indian Statistical Institute, Kolkata



Collaborators

This research is in collaboration with Mr. Chayan Maitra from Machine Intelligence Unit, Indian Statistical Institute, Kolkata.



Citation

A more detailed version of this work has been published in Neural Networks. Maitra, C. and De, R.K., 2025. G-NeuroDAVIS: A generative model for data visualization through a generalized embedding. Neural Networks, p.107948.



Summary

Visualizing high-dimensional datasets has been a longstanding challenge. Several methods have been proposed for this purpose, but they have yet to generate a generalized embedding that not only reveals the hidden patterns present in the data but also generates realistic high-dimensional samples from it. Motivated by this aspect, a novel generative model, called G-NeuroDAVIS, has been developed, which is capable of visualizing high-dimensional data through a generalized embedding and thereby generating new samples. The model leverages advanced generative techniques to produce high-quality embedding that captures the underlying structure of the data more effectively compared with the existing methods. G-NeuroDAVIS can be trained in both supervised and unsupervised settings. We have rigorously evaluated our model through a series of experiments, demonstrating superior performance in several downstream tasks, which highlights the effectiveness of the learned representations. G-NeuroDAVIS has outperformed Variational

Autoencoder (VAE) significantly in terms of embedding quality and downstream tasks like classification. Moreover, the superior sample generation capability of G-NeuroDAVIS has been demonstrated against VAE, Deep Convolutional Generative Adversarial Network (DCGAN), Denoising Diffusion Probabilistic Models (DDPM), and Autoencoder (AE)-guided Real-valued Non-Volume Preserving (RealNVP). These results highlight the efficacy of G-NeuroDAVIS to serve as a robust tool in various applications that demand high-quality data generation and representation learning.



Methodology

Traditional methodology, like t-SNE (t-distributed Stochastic Neighbor Embedding) and PCA (Principal Component Analysis), lack generative capability and often distort global data structure. VAEs (Variational Autoencoders used to create new data similar to the input it was trained on) generate blurry, low-fidelity outputs, especially for complex datasets like faces or objects. This research asset outperforms other generative models for generating data from target sets such as DCGAN (Deep Convolutional Generative Adversarial Network) and DDPM (Denoising Diffusion Probabilistic Model).

G-NeuroDAVIS takes an identity matrix as input and reconstructs the original data through a few hidden layers. During this reconstruction, it learns a close approximation of the actual data distribution using suitable

parameters in the latent space. Once the latent distribution is learned, several samples in the high-dimensional space are generated by sampling from this distribution. Effectiveness of this generation process demonstrates generalizability of the latent space, implying that it can distinguish samples based on some specific attributes. A generalized latent space allows a smooth transition from one cluster to another, thus reflecting an effective interpolation capability of the same. Although G-NeuroDAVIS can generate high-dimensional samples successfully, it lacks control over the output as each generated sample is independent and not influenced by any specific conditions or prior generations. As a result, the model cannot be controlled to produce samples with desired properties or characteristics to a particular class, limiting its applicability in scenarios where conditional generation is required. In order to tackle this issue, the research asset has modified the input using the label information and learned a conditional distribution at the latent space in a supervised fashion. For incorporating the label information, instead of using an Identity matrix, a one-shot encoder matrix (order being dependent on the number of samples and classes) is considered. This input ensures that all the samples belonging to a particular class are projected to the latent space through specific weight parameters. This supervised model is efficient enough to generate samples based on a specific condition. Thus, G-NeuroDAVIS is a model that can visualize data through a generalized embedding, produce samples in both conditional (supervised) and unconditional (unsupervised) scenarios, and can handle high-dimensional complex datasets.



Architecture: G-NeuroDAVIS includes an input layer that takes an identity matrix as input. Parametric layer controls latent distribution ($\sim N(0,1)$). Latent layer takes a sample from the latent distribution, and produces generalized embedding of data after successful training. Hidden layers capture the non-linear relationships present in the data and while the reconstruction layer reconstructs the data.



Learning: Learning is done through two losses - for realistic reconstruction (MSE) and for keeping latent distribution $\sim N(0,1)$ (K-L Divergence). After training, generalized embedding is obtained from latent layers; and samples are generated from embedding using hidden and reconstruction layers. Thus, the loss function is given by

$$\mathcal{L}_{G-NeuroDAVIS} = \frac{1}{n} \sum_{i=1}^n \|\mathbf{x}_i - \tilde{\mathbf{x}}_i\|_2^2 + \beta \sum_{i=1}^n KLD\left(\mathcal{N}(\mathbf{h}_i^{(\mu)}, \exp(\frac{\mathbf{h}_i^{(\sigma)}}{2})) \parallel \mathcal{N}(\mathbf{0}, \mathbf{I})\right)$$



Training: For conditional data generation, the input layer takes a sparse matrix ($n \times c$) as input, n is number of samples and c is number of classes present. Parametric layer controls class wise latent distribution. Latent layer takes a sample from the latent distribution, and produces a generalized embedding of the data after a successful training.



Inference: A one-hot vector is provided as input, with the “1” placed at the index of the target class. The trained parameters map this vector through the latent space, producing an embedding located near the distribution of the chosen class. The reconstruction layers then transform the latent embedding back to the original data space via the hidden layers, generating a sample corresponding to the specified class.



Novelty

By combining data visualization and sample generation in a single flexible model, it helps make high-dimensional information contained in images, health records, or

user behavior, easier to interpret and use responsibly. For instance, with better ways to visualize patient data or generate realistic samples, tools like G NeuroDAVIS can support doctors in making quicker, more informed decisions, especially when working with limited or sensitive data.



Validation

The suggested approach of G-NeuroDAVIS is tested and validated for producing generalized embedding, and generating meaningful, diverse and realistic samples. It works in both supervised and unsupervised modes, and enables conditional generation. It is tested for better separation of clusters in the latent space of G-NeuroDAVIS. It has demonstrated effective performance in generating sharper and clearer digits compared to VAE, and also superior classification performance of G-NeuroDAVIS generated embedding over VAE. Conditional data generation capability of G-NeuroDAVIS is realistic in nature. The embedding shows clear separation of classes. G-NeuroDAVIS generated samples are authentic and diverse according to improved precision and recall scores.

From a performance perspective, this approach has similar visualization to VAE, but better classification and interpolation capability than VAE, DCGAN, and DDPM; comparable to AE-guided RealNVP. G-NeuroDAVIS works with fewer parameters than other state-of-the-art models.



Impact

G NeuroDAVIS has the potential to make meaningful contributions to how we understand and interact with complex data in real life. As it works well even when labeled data is scarce, it could be useful in areas that do not have access to large, curated datasets. The model also encourages greater transparency in machine learning. Instead of acting like a “black box”, it allows people to see how the model is understanding and grouping information, which can build more trust in automated systems. Finally, by helping researchers and students better visualize abstract patterns, G NeuroDAVIS can support education and open up AI development to more people — not just those with deep technical backgrounds.



Future Scope

Future research scope includes reduction in space complexity, developing robust hyperparameter tuning strategies, exploring non-Gaussian latent distributions for embeddings, extending to more advanced conditional generation frameworks, improving smoothness and continuity in latent transitions, and applying it for data imputation.

ZIF (Zero Incident Framework) for Autonomous Incident Management in Service Delivery



Presenter

Mr. Balaji Uppili, CBO - India & UK, Neurealm



Collaborators

This pioneering research and product development is a crucial collaboration between Neurealm and the esteemed IIT-Madras. The core team driving this innovation includes Naveenraj, Aravind Balaji, Dhanabalan Chellakannu, Vijay Arvind Jawahar, Suresh Kumar Ramasamy et al. This product offering is already deployed and live at key customer data centers globally.



Summary

The core challenge in modern IT is business disruption caused by outages, typically

met with reactive responses that incur high cost, low efficiency, and lost customer satisfaction. The imperative is clear: IT infrastructure services must transition to a Predictive, Proactive, and Zero Downtime model with autonomous correction. ZIF is Neurealm's patented method and system for determining the comprehensive health of assets in an information technology (IT) environment, including web applications, servers, networks - essentially anything with an IP address - as well as all applications and their associated data, such as server and audit logs.

This invention aims to provide a comprehensive and accurate real-time analysis of application health. Automaticity in IT infrastructure management (AIOps) presents both a huge technical challenge and a powerful business opportunity for achieving zero downtime. This asset directly addresses the complete problem, spanning the entire lifecycle from observability and data collection to data normalization, analysis, recommendation, and auto remediation.



Methodology

ZIF's core methodology establishes a proactive, data-driven approach:

Receiving Data

Data is received for various parameters (exceptions, network anomalies, resource performance, and user experience) from network nodes including web servers, application servers, database servers, and end-user devices.

Determining Scores

Scores are computed for these parameters - for example, user experience scores (based on response time and 'wow-woe' counts), exception scores, network anomaly scores, and resource performance scores.

Assigning Ratings

Each score is compared against a multi-rating scale with multiple threshold levels (for example, 'excellent', 'very good', 'must improve', 'unacceptable') to assign a parameter rating.

Determining Final Ratings

Final ratings are determined for each category (exceptions, network anomalies, resource performance, user experience) by aggregating pluralities of parameters and combining individual ratings using predetermined weights.

Generating a Scorecard

A real-time Application Health Index (AHI) or scorecard is generated based on these final ratings, providing an accurate, 360-degree indication of the application's health.

Universal connector

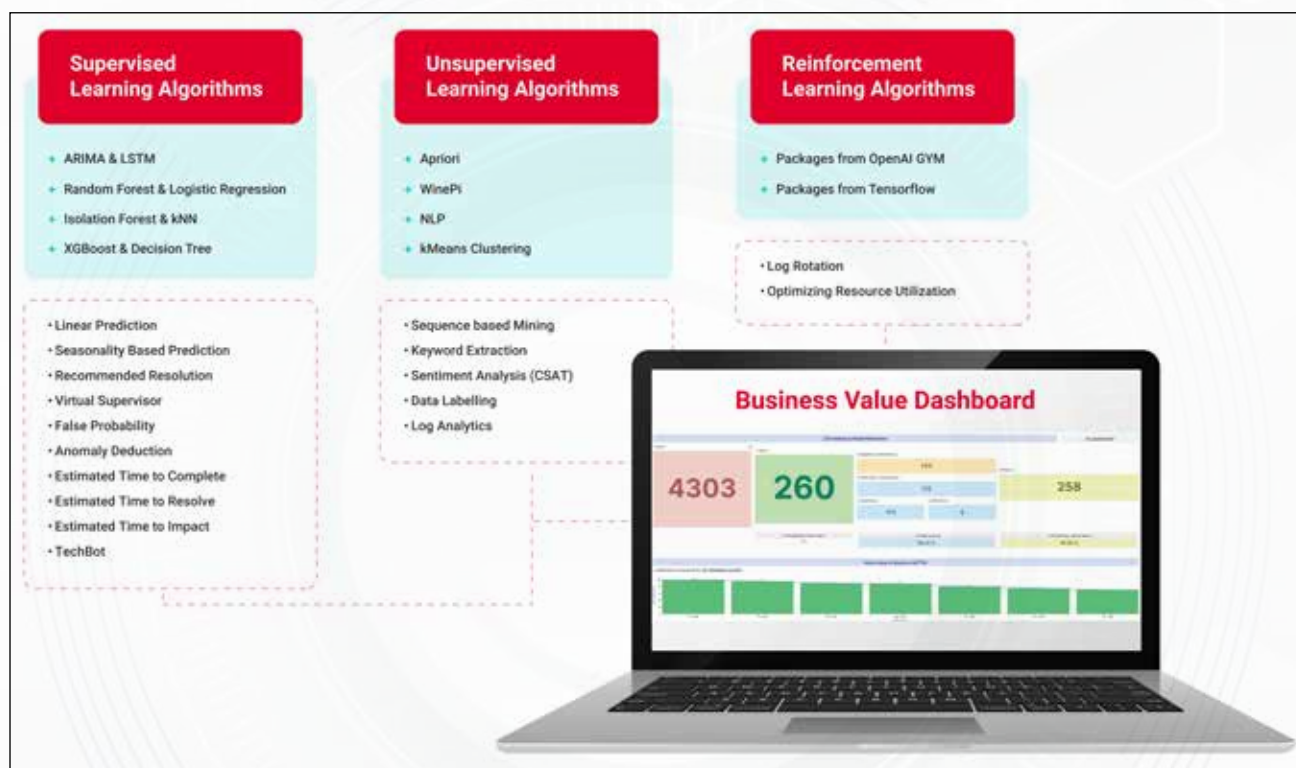
This acts as a global bridge through which data from multiple sources can be fed into the platform. The data and insight from multiple sources are parsed through this universal connector.

AI Models & Automation

Various AI models are used within the platform; the model details are captured in the image below. The platform utilizes AI techniques and ML algorithms, specifically supervised learning, unsupervised learning, and reinforcement learning. They are utilized for prediction, anomaly detection, arriving at dynamic thresholds, and also to do capacity forecasting. There are 300+ automation bots within the platform where AI plays a major role. These bots are further extensible through a workflow-driven approach.

The system typically includes a data reception module, a score computation module, a rating module, and an application health module. These modules work together to receive data, compute scores, assign ratings, and generate the final application health index. The system can communicate with various network devices such as web servers, application servers, database servers, and end-user devices over a network.

AI Algorithms

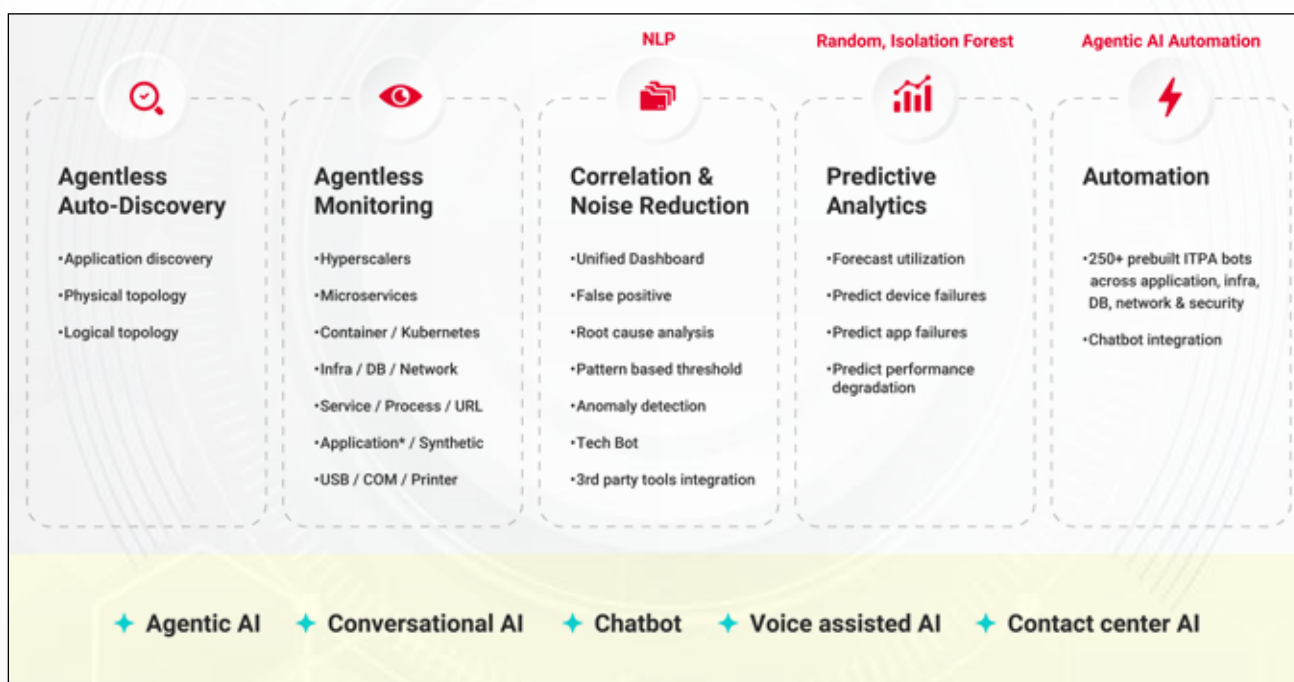


The front end includes agentless auto discovery, agentless monitoring, auto correlation and noise reduction, predictive analytics and remedial automation, all powered through Agentic AI with an AI Chatbot. ZIF can also be integrated with third party monitoring and data collection or observability tools for seamless data center operations.



Novelty

The USP of ZIF is that it not only provides auto remediations and recommendations but also provides RCA and causal analysis. ZIF incorporates novel and unique features such as Agentless auto discovery, Agentless



monitoring, Multi variate correlation and noise suppression, predictive analytics for incident predictions, and remediation recommendations and service automation. ZIF also supports AI based conversational chatbot and contact center and it incorporates a universal connector workflow which can collect monitoring data even from third party tools.



Validation

ZIF is a highly matured innovation currently positioned at TRL Level 5, and deployed at client data centers across the world. It has successfully transitioned from a research asset into a live, production-grade platform by systematically moving beyond theoretical concepts to a scalable, user-centric solution.



Impact

Deploying ZIF accelerates business outcomes across three dimensions:



Efficiency & IT Outcomes

- Improved cost-to-income ratio
- Faster transaction processing time
- Increased customer satisfaction (CSAT)
- Improved employee productivity



Customer Experience & Risk

- Improved accessibility and reduced churn rate
- Better identification and prevention of fraudulent activities
- Improved profitability management through intelligent IT cost management



Social Impact

- Reduced economic disruption through resilient infrastructure
- Improved quality of life with reliable access to essential services (for example, healthcare, finance, transportation)



Future Scope

Future scope involves improving predictions with reduced false positives and improving insight generation. We will also focus on enhancing auto-remediation

(not just recommendation) with improved precision and recall for AI models. More research is needed on how GenAI assist can automate any Level-1/2/3 job and provide multi-variate correlation across numerous distinct parameters to improve prediction accuracy.



MOYA - Meta

Orchestrator of your Agents



Presenter

Dr. Karthik Vaidhyanathan, Assistant Professor, IIIT Hyderabad



Collaborators

This research is in collaboration with Karthik Vaidhyanathan, Rudra Dhar, Adyansh Kakran, Sreemaee Akshathala, Shrikara Arun, Amey Karan from IIIT Hyderabad and Kannan Parthasarathy, Venkat Krishnamachari, Basil Muhammed, Sumant Dubey, and Mohan Veerubhotla from MontyCloud Inc, Bangalore.



Summary

Meta Orchestrator of Your Agents (MOYA) is a modular, multi-agent framework for Autonomous CloudOps. It provides an agent marketplace, a tool registry, and an orchestration engine to build, integrate, and

manage CloudOps agents. First engineered multi-agent framework for CloudOps, which is typically complex, error-prone, and resource-intensive. MOYA is also extended to create an ecosystem of reusable agents for CloudOps automation, and it enables transparency, peer validation, and faster innovation cycles and supports plug-and-play agents for new enterprise needs, adaptable to multiple cloud vendors. Although MOYA was developed for autonomous CloudOps, it has been shown to generalize to different domains as a generic, lightweight, agentic AI framework.



Methodology

Monolithic LLM/RAG is not scalable as it is hard to update, brittle with workflows and slows down at enterprise scale. Security, reporting, ticketing, optimization, etc each requires a specialized capability. Also, bloated “generalist” systems fail to adapt at enterprise scale.

Cloud Operations (CloudOps) is a rapidly growing field focused on the automated management and optimization of cloud

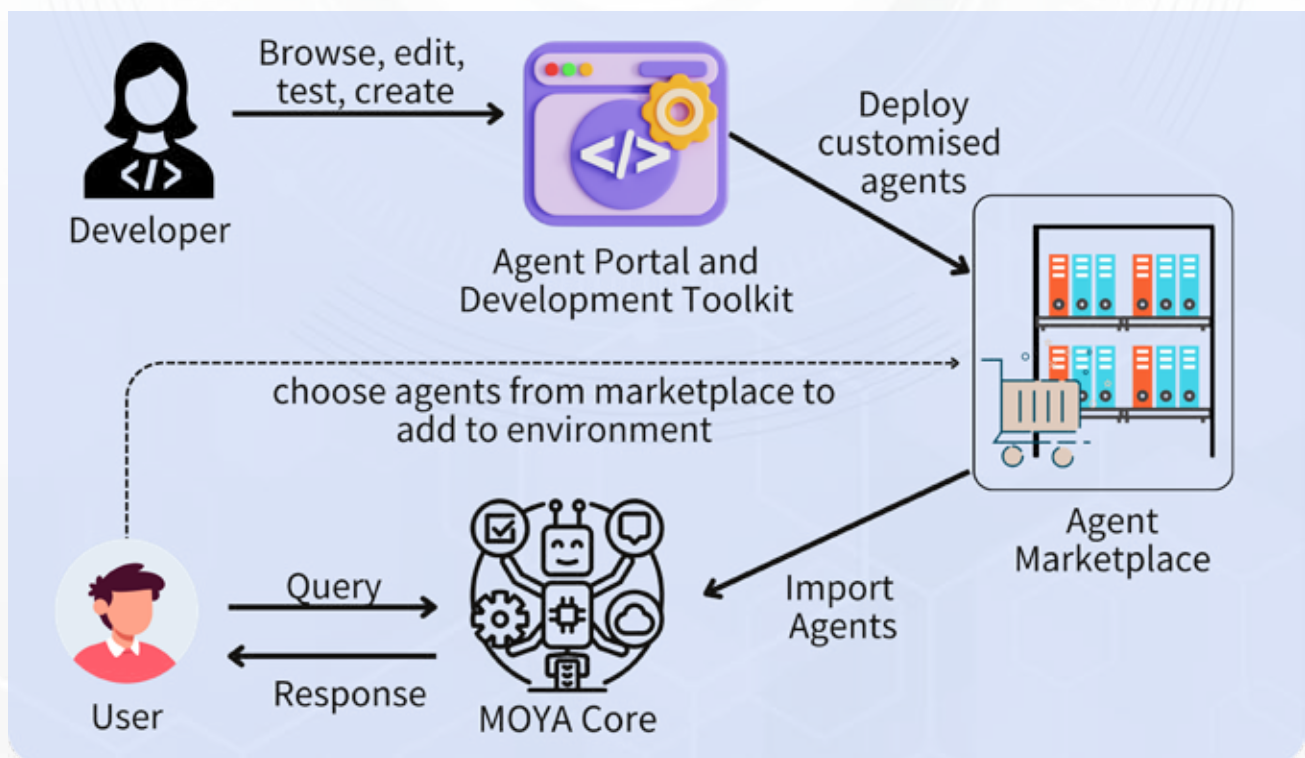
infrastructure, which is essential for organizations navigating increasingly complex cloud environments.

MOYA was initially designed for Autonomous CloudOps in Collaboration with MontyCloud, but through its open-source release and a community student hackathon on the IIIT-H campus, it has already shown applicability across multiple domains. Although built for CloudOps, its use has extended more broadly to different domains also highlighting its potential to serve as a training and education framework for Agentic AI. By lowering entry barriers, MOYA democratizes innovation, fosters skill development, and accelerates the adoption of sustainable AI practices. Its impact spans both technological efficiency and capacity-building.

Developing a GenAI-based solution for autonomous CloudOps presents various

challenges such as diverse data sources, orchestration of multiple processes and handling complex workflows to automate routine tasks. This led to the innovation called MOYA, a multi-agent framework that leverages GenAI and balances autonomy with the necessary human control. This framework integrates various internal and external systems and is optimized for factors like task orchestration, security, and error mitigation while producing accurate, reliable, and relevant insights by utilizing Retrieval Augmented Generation (RAG).

Fundamental idea here is to support custom agents, in-house agents, tools, etc and orchestrate with guardrails. The meta orchestrator core engine consists of Request handler, Orchestrator, Agent selection, LLM/ Non-LLM processor, Monitoring, Guardrails and Agent invocation from Agent library with lifecycle management.





Novelty

MOYA is a framework for managing & orchestrating agents. Some features: Coordinates across any agent (internal, external, or third-party), balances automation with human oversight, and domain-driven design principles make it modular, extensible and secure.



Impact

Compared to monolithic RAG, the MOYA framework has less hallucination as it understands the tasks better, has parallel agents for faster results, and produces reliable results with efficient token usage.



Validation

Evaluations of this multi-agent system, with the help of practitioners as well as using automated checks, demonstrate enhanced accuracy, responsiveness, and effectiveness over non-agentic approaches across complex workflows. Here is a chart on the combination of automated and manual evaluations with ground truth of 260 prompts and responses.



Future Scope

More research and innovation are needed to further extend MOYA's capabilities with different types of memories, interoperability for tools and multi-agent interaction with support for protocols like MCP, A2A, etc. Further, Agentic AI in general provides a lot of promise, such as for software modernization and support for automation of various traditional software engineering practices. With the increasing concern on sustainability, there is also a big need to develop domain-specific Small Language Models (SLMs) that can power the agents, and there is also a need for better processes to architect/engineer systems around AI agents. Agentic AI offers promise to reimagine autonomy, sustainability and intelligence at scale.

Approach	Rouge-1	BLEU	Meteor	BERT score		
				Precision	Recall	f1
Monolith	0.321	0.102	0.265	0.854	0.834	0.843
MOYA	0.448	0.221	0.423	0.867	0.869	0.868

IBM Granite

Time Series Foundation Models



Presenter

Dr. Pankaj Dayama, Senior Technical Staff Member & Senior Manager, IBM Research - India



Collaborators

This research asset is in collaboration with other IBM Researchers including Vijay Ekambaram, Arindam Jati, Sumanta Mukherjee, Subodh Kumar, Chandra Reddy, Wesley Gifford & Jayant Kalagnanam.



Summary

IBM Granite TinyTimeMixers (TTMs) and IBM Granite TSPulse are part of the IBM Granite Time Series Foundation Model (TSFM) family developed by IBM Research. TTMs are compact pre-trained models for multivariate time-series forecasting, with sizes starting at just 1M parameters. They

are “tiny” pre-trained models designed specifically for time-series forecasting, and the research paper on TTM was accepted at NeurIPS 2024. TSPulse models are ultra-compact pre-trained models ($\approx 1M$ parameters) for time-series analysis, enabling GPU-free inference. They are versatile across multiple tasks such as classification, anomaly detection (AD), imputation, and similarity search. Together, TTMs and TSPulse deliver state-of-the-art accuracy with extremely lightweight architectures suitable for enterprise-scale deployment.



Methodology

Time-series analysis is fundamental to enterprise operations, powering use cases ranging from industrial equipment monitoring to sales and inventory forecasting. Industrial sensor data from plants and equipment provide critical signals that help prevent failures, minimize shutdowns, and support optimal control decisions. Similarly, analyzing historical operational data enables improved forecasting and planning across business processes.

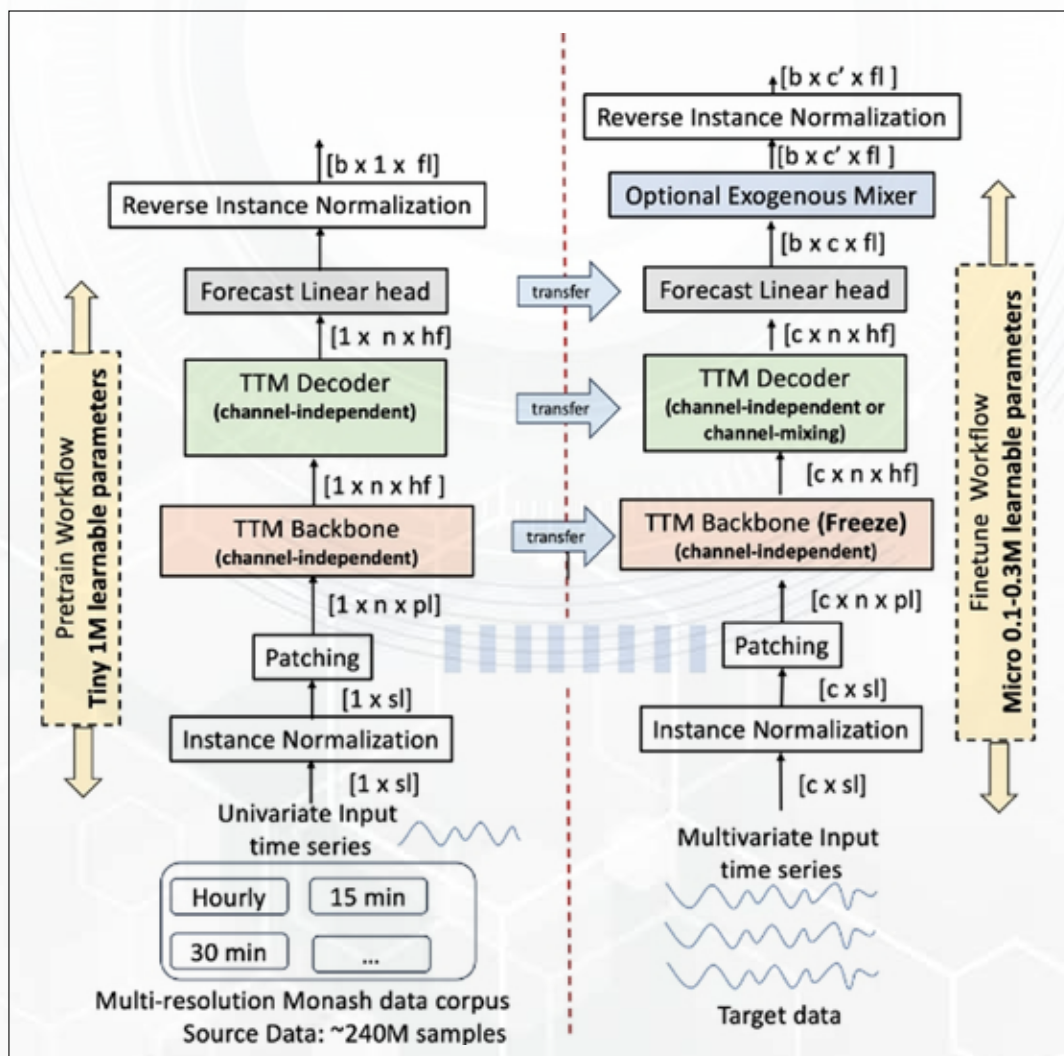
Multivariate time series involve multiple variables that evolve together over time, where each variable may depend on both its past values and the values of other correlated variables. Building foundation models for such data introduces several challenges:

- **Complex correlations:** Time dependencies occur both within each series and across multiple channels.
- **Domain variability:** Time-series characteristics (sampling rates, numeric scales, seasonality, drift, noise profiles) vary widely across industries.

- **Data scarcity:** Public datasets are limited, often small, and rarely aligned with the target domain.
- **Deployment constraints:** Enterprise adoption requires accurate zero/one-shot multivariate forecasting, GPU-free lightweight models, and support for exogenous variables and cross-channel fine-tuning.

The Granite TSFM family addresses these challenges through pretraining strategies and architectures tailored for real-world, multivariate industrial and enterprise time-series data.

Figure 1: Granite TTM Model





Novelty

The Granite Time Series Foundation Models (TSPulse and TTM) introduce breakthroughs in accuracy, efficiency, and deployability for multivariate time-series analysis.

TSPulse

TSPulse outperforms state-of-the-art benchmarks while being exceptionally lightweight and resource efficient. Key innovations include:

- 01 +5–16% accuracy gains on UEA classification benchmarks
- 02 Over 50% improvement in zero-shot imputation
- 03 100× smaller architecture than leading SOTA Transformer models (~1M parameters)
- 04 GPU-free inference and fast CPU-based fine-tuning
- 05 Extremely low memory footprint, eliminating out-of-memory failures even on constrained devices
- 06 Ideal for the Indian ecosystem, where low-cost, real-time, compute-efficient AI is critical

TinyTimeMixers (TTM)

TTMs are compact (~1M parameters) foundation models for multivariate forecasting and zero/one-shot prediction. Notable innovations include:

- 01 High forecasting accuracy despite tiny model size
- 02 Multi-resolution temporal mixing enabling strong performance across diverse time scales
- 03 Support for exogenous variables and cross-channel correlations
- 04 Rapid fine-tuning for domain-specific industrial datasets
- 05 GPU-free and edge-friendly inference, enabling deployment directly on production systems
- 06 Accepted at NeurIPS 2024, highlighting novelty and scientific contribution

Together, TSPulse and TTM redefine the design space of foundation models, delivering SOTA performance with ultra-small models, enabling real-time adoption across enterprise and industrial environments.



Validation

These models are already released at Hugging Face with over 20M downloads

already. The USP for this is extremely small models (10-100X smaller), GPU-Free Inference & tuned for real-time industrial use-cases where both latency and accuracy matters together.

Ref: <https://huggingface.co/ibm-granite/granite-timeseries-tspulse-r1> & <https://huggingface.co/ibm-granite/granite-timeseries-ttm-r2>



Impact

These models are already infused into IBM products like Storage Insights and watsonx.ai and are being used for client-specific fine-tuning to deliver measurable value, especially in the process industry. By being

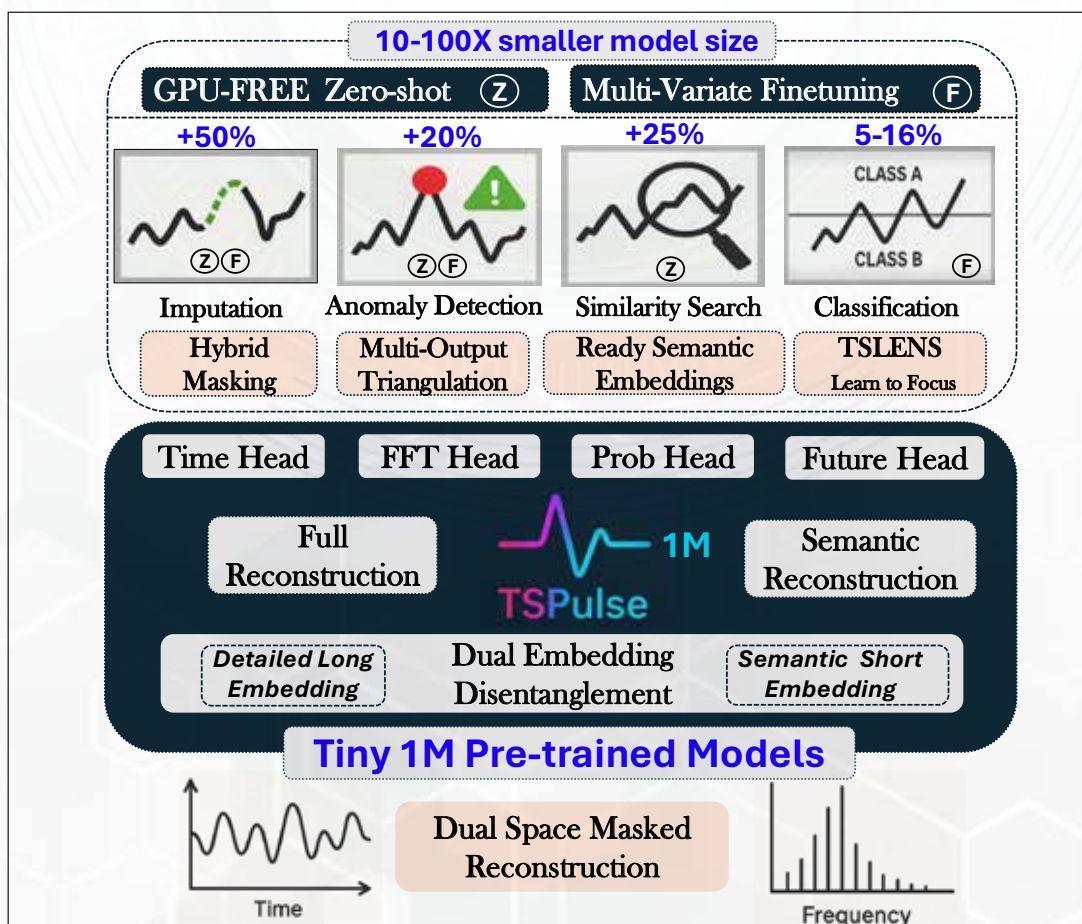
extremely lightweight and resource-efficient, these models can run on low-cost hardware without GPUs, making AI-driven decision-making accessible and scalable across industries. This promotes efficient resource use, improved operational reliability, and reduced energy consumption, contributing to sustainability goals and enabling organizations to achieve more with fewer computational resources.



Future Scope

Domain specific tuning of TTM/ TSPulse for domains such as Energy, Finance, IT Operations, Manufacturing.

Figure 2: Granite TSPulse Model



Warehouse Autonomous Sorter and Picker (WASP)



Presenter

Dr. Sachit Rao, Associate Professor, IIIT Bangalore



Collaborators

Mr. Karthikeyan Sundaram (CTO) & Mr Anjan Kumar (CEO), Greendzine Technologies Pvt Ltd, Bangalore



Summary

A plug-and-play Autonomous Mobile Robot (AMR) solution that can seamlessly integrate with existing warehouse infrastructure. Picking/Put-away is one of the most important processes in any warehouse, especially e-commerce. In India, also globally, this is manual and hence labour intensive; human performance decides

throughput and efficiency. Automated solutions for pallets and large items exist (ASRS, Goods-to-person) but these solutions pose several challenges, such as: they are not designed to work together with human workers; they are not plug-and-play; they require extensive modifications to the warehouse; and they cannot handle the wide variety (in size, texture, hardness etc.) of items in a warehouse. An AMR that can work with humans, handle diverse objects, and can navigate safely and autonomously in dynamically changing environments is the need.



Methodology

The solution comprises of an Autonomous Mobile Robot (AMR) equipped with sensors/compute for safe navigation in dynamically changing environments. A Universal Jamming Gripper (developed by the collaborators) connected to a 6-DoF industrial manipulator, which in turn is mounted on the AMR, is used for picking diverse objects. By design, the Jamming

Gripper can adapt to objects of different geometries and textures.

The AMR comprises of independently controlled BLDC motors equipped with Hall sensors and encoders; for sensing, it has a 2-D LIDAR scanner, a 3-D depth camera, and ultrasonic sensors; the data from these sensors are deployed in algorithms that are executed on an on-board CPU-GPU computing system.

The AMR follows a path planning algorithm that dynamically adjusts paths and ensures collision avoidance based on real-time changes in the environment and workload. The AMR is driven by a pair of hub BLDC motors mounted directly on the wheels; the inputs are provided by CAN protocols.

The AMR can carry upto 110 kg payloads and has a pulling load of 250-300 kg. The system is also accompanied by a web application designed for customers to monitor and control their deployments. The preliminary functionalities for this application have been identified, and the UI/UX prototype is currently in development.

For object manipulation, it is first identified using computer-vision techniques, such as the SIFT algorithm; this algorithm is used to match the desired item to be picked from a crowded shelf in the warehouse. After object identification, using outputs of a depth camera, which is also mounted on the manipulator along with the Jamming Gripper, the location of the object is determined and next, the on-board manipulator is commanded to pick the object from this location.

Autonomous Collision-free Navigation is achieved by providing the AMR a map of its working environment. Based on this map, algorithms that can find a path between 2 points on the map are executed. Prior to movement, the AMR first localizes itself (finds its place in the map) using LIDAR/camera data, and post this operation, movement is generated by providing appropriate commands to the motors so that the AMR traces the designated path. Sensors on the robot ensure collisions are avoided; a new path is found if the original path cannot be traced.



Novelty

A mobile manipulator with a gripper that can manipulate diverse objects. Autonomous and safe navigation in crowded environments. Designed to handle large payloads (>100 kg), move at high speeds (< 10km/hr), and handle variations in the warehouse floor



(upto 3 degrees of inclination). Compliant with ISO and ANSI standards for functional safety and navigation.



Impact

The AMR system is designed to work with human operators in warehouses. It is expected to reduce the fatigue experienced by pickers. This in turn can increase their productivity and also boost employment of women in such physically demanding tasks.

The development of this project, which is an industry-university collaboration result, has also trained several people in the domains of engineering and other deep-tech areas.



Future Scope

Future scope includes iterate engineering, optimize paths for multi-agent case; currently heuristics are applied and implement visual SLAM.



IBM Geo-spatial Foundation Model

Prithvi - IBM Geo-spatial Foundation Model



Presenter

Dr. Manikandan Padmanaban, Senior Research Engineer, IBM Research Lab India



Collaborators

This research is in collaboration with Jagabondhu Hazra, Kamal Das, Amith Singhee, Paolo Fraccaro, Anne Jones, Johannes Jakubik, Campbell Watson, Juan Bernabe Moreno from IBM Research.



Summary

Prithvi is a family of geospatial foundation models developed collaboratively by IBM Research and NASA to advance Earth observation (EO) applications. Prithvi represents a significant step toward universal geospatial foundation models, combining large-scale self-supervised training on Harmonized Landsat and Sentinel 2 (HLS)

data, temporal awareness, and global coverage. By leveraging transformer-based masked autoencoder architectures, Prithvi provides robust representations that are effective for downstream applications in climate science, disaster management, and sustainable development.



Methodology

Prithvi represents IBM's Geospatial foundation models such as Earth Observation Foundation Model (Prithvi-EO) and Weather and Climate Foundation Model (Prithvi-WxC). Prithvi-EO is first-of-a-kind foundation model adapting vision transformers for satellite data built by NASA and IBM.

Prithvi builds upon the Vision Transformer (ViT) and Masked Autoencoder (MAE) paradigms. In pretraining, random portions of image patches are masked, and the model learns to reconstruct them from context, forcing it to capture both spatial and temporal dependencies. The architecture integrates temporal embeddings, enabling effective modeling of time-series EO data,

a critical feature for monitoring land-use change, crop growth, or disaster impacts.

The latest version, Prithvi-EO-2.0, was trained on 4.2 million time series tiles of globally distributed HLS imagery. This dataset spans multiple ecoregions and climate zones, ensuring that the model is not biased toward specific geographies. The models range in scale from ~100M to over 600M parameters, balancing performance with computational feasibility. Training emphasizes global diversity, which allows Prithvi to generalize effectively across different continents, land types, and seasonal patterns.

Prithvi demonstrates state-of-the-art results on GEO-Bench, a standardized benchmark suite for geospatial foundation models. Compared to earlier models, Prithvi-EO-2.0 shows up to an 8% improvement in average accuracy across tasks such as land

cover classification, flood detection, and vegetation monitoring. Importantly, the model achieves competitive results even in low-shot scenarios, where labeled data is scarce—a major advantage in remote sensing, where annotations are expensive and limited. Due to its flexible representations, Prithvi can be fine-tuned for a wide variety of tasks such as rapid flood mapping, wildfire burn scar segmentation, damage assessment, crop type classification, yield prediction, irrigation monitoring, vegetation dynamics, deforestation tracking, and urban expansion.

Prithvi's weights and fine-tuning pipelines are openly available on Hugging Face, lowering barriers to adoption. The ecosystem includes prebuilt training workflows, enabling researchers, NGOs, and enterprises to adapt the models to their needs. IBM has also begun integrating Prithvi into its enterprise

Model Architecture

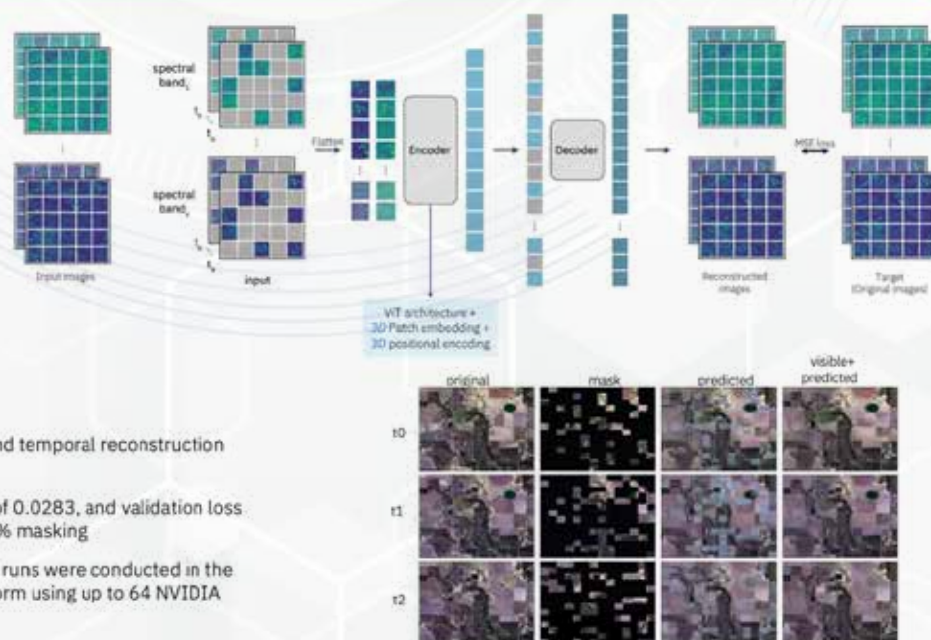
- Pre-training task: reconstruct *masked* patches → target = original data.

- MSE loss on *masked* patches.

Encoder → Vision transformer (ViT/Swin) for multispectral 3D data.

- 3D patch embeddings
- 3D positional encoding

- Excellent spatial and temporal reconstruction performances
- Training MSE loss of 0.0283, and validation loss of 0.0364 with 75 % masking
- All the pre-training runs were conducted in the IBM watsonx platform using up to 64 NVIDIA A100 GPUs.



products, enhancing geospatial analytics within business and climate resilience solutions. Pre training data includes data from ISRO EOS-04 and RESOURCESAT-2/2A satellites with 18m , 56m, 24m and 5m accuracy respectively with 2.5 to 3.8TB data and ESA (European Space Agency) Sentinel-1/2 satellites with 10m accuracy for 5 to 20TB data. The model architecture includes pre-training task to reconstruct masked patches and MSE loss on masked patches and pass it to vision transformer ViT & Swin (Shifted Window Transformer) encoder for multispectral 3D data with 3D patch embeddings and 3D positional encoding. The model performed excellent with training MSE (mean square error) loss of 0.028 and validation loss of 0.036 with 75% masking.



Novelty

This is pioneering country-specific Geospatial Foundation Model developed using native datasets from Indian satellites (IRS: EOS-4, Resourcesat-2/2A) and European Space Agency satellites (Sentinel-1, Sentinel-2) with systematic integration of geospatial variables with spatial distribution constraints, crucial for applications like environmental monitoring, precision farming, and other Earth observation tasks. These models are trained on the multispectral and microwave datasets, enabling robust performance under diverse weather conditions. These models have enhanced Reconstruction Efficiency as it leverages spatial and

spectral masking techniques during training to boost the model's reconstruction capabilities. Superior performance via spatiotemporal encoding through spatial and temporal encoding to achieve higher accuracy and reliability across tasks.



Validation

The Prithvi-EO has been validated on various tasks and use cases such as segmentation (flood detection, fire scar detection), regression (Biomass estimation, urban heat islands) and multi-class segmentation (crop classification). Whereas Prithvi-WxC which has applications in understanding of weather and extreme events on both shorter (e.g., 10 days) and longer time horizons (e.g., 100 years in the future) and can account for global long-term effects like El Niño has been validated for HiRes precip nowcasting, weather forecasting, climate modelling, hurricane tracking and gravity waves parameters.

The model has been tested for applications such as Flood Mapping using SAR (Synthetic Aperture Radar), Crop mapping to help crop estimation and this asset has been released as GeoSpatial toolkit called “TerraTorch” built upon PyTorch and TeorchGeo (<https://github.com/IBM/terratorch>). At the time when tested, On GEO-Bench (a benchmark suite for Earth observation), Prithvi outperforms its predecessor by ~8%, and also outperforms a number of other existing geospatial foundation models.



Impact

Trained on global satellite imagery, Prithvi can detect floods, wildfires, droughts, and land-use changes with high accuracy, enabling faster and more effective disaster response. For example, emergency teams can use Prithvi-based models to map flood-affected areas in near real time, helping direct aid and reduce loss of life. For instance, the automated AI based flood detection has 90%+accuracy and is capable of flood mapping in all weather condition using SAR detecting floods at national scale.



Future Scope

These foundational models can be evolved and fine-tuned for other data and info (e.g., fine-tuning with S1 radar data to detect flood on cloudy days). For instance, Prithvi-WxC can be extended for versatile and generalizable standalone models that can be combined to further unlock new use cases for earth weather, hurricane and climate.



Productization

From Research to Product

- First-of-its-kind Geo-spatial foundation model on open data set
- Open-sourced on Hugging Face for community adoption and validation.
- Integrated into IBM offerings and adopted by various governments.

Scalability

- Pre-trained on global satellite data
- Minimal fine-tuning needed across domain and applications
- Already finetuned, validated and open sourced for applications like crop identification, flood, burn scar, land slide detection, above ground biomass, land use land cover, etc

Business & Social Impact

- Accelerated product development and innovations in Earth Observations industry
- The model has been adopted by Govt of USA, UK and Kenya.
- NASA Agency Group Achievement Award and AGU 2025 community award
- Hackathon(IBM & IISc) specific to Indian socio-economic challenges supported by MeiT and IndiaAI

Information Theoretic Approach for Scaling Speech AI - When Less Becomes More



Presenter

Mr. Sourav Banerjee, Founder and CTO
Shunya Labs and United We Care



Collaborators

This asset is in collaboration with Abeer
Sehrawat and Ayush Kumar from Shunya
Labs.



Summary

Pingala V1 is Shunya Labs' Automatic Speech Recognition foundation model which has achieved 3.10% WER, putting it at #1 on the Open ASR leaderboard, with coverage for 200+ languages and <100 ms latency, trained on 1000 times less GPU hours than standard SOTA models.

The problem statement here is that the state-of-the-art ASR today is built on data bloat: millions of hours of loosely filtered audio,

trained for weeks on hundreds of GPUs. This creates steep access barriers, locking out academia, startups, and even whole geographies while carrying an unnecessary environmental cost. The field has become quality-blind, optimising for dataset size rather than information density, so models waste compute on redundant speech and overfit dominant patterns. Multilingual performance also suffers because labeled audio in low-resource languages is scarce and expensive, and models remain fragile across domains, with error rates spiking on meetings, earnings calls, and other real use cases. Some SOTA models, for example, have ~10% WER on meetings/earnings.

Our intervention in the field involves replacing the assumption of more data with the principle of more information. We achieved 3.10% WER by measuring Shannon-entropy for each dataset and creating an algorithm to train our model with increasing levels of surprising information.



Methodology

Our training methodology assigns each audio sample x_i a composite information score

built from four entropy terms—acoustic, phonetic, linguistic, and contextual. For a sample with features $(x_i(a), x_i(p), x_i(l), x_i(c))$, and transcription y_i , we define

$$H_{\text{comp}}(x_i) = \sum_{k \in \{a, p, l, c\}} \alpha_k H_k(x_i) + \beta I(x_i, y_i),$$

where $I(x_i, y_i)$ is the mutual information between input and label, and α_k, β control the contribution of each difficulty dimension.

Intuitively, $H_{\text{comp}}(x_i)$ is a single scalar “difficulty” or information-density score: it is larger when the audio is acoustically complex, phonetically diverse, linguistically/ contextually rich, and strongly predictive of its label.

Based on H_{comp} , we update an adaptive threshold τ_k at pruning round k ,

$$\tau_k = \tau_0 e^{-\gamma k} \left(1 + \varepsilon \sin \frac{2\pi k}{\rho} \right),$$

and retain each sample with probability

$$P_{\text{retain}}(x_i, k) = \frac{1}{1 + \exp(-\sigma(H_{\text{comp}}(x_i) - \tau_k))}.$$

The exponential term makes the curriculum progressively more selective, the sinusoidal perturbation preserves diversity, and the sigmoid implements soft selection.

Data selection and model training are coupled via

$$\mathcal{L} = \max_{D_k} \left[\text{JS}(D_{k-1}, D_k) - \lambda_1 \frac{|D_k|}{|D_0|} - \lambda_2 \mathcal{L}_{\text{task}}(\theta_k) \right],$$

trading off diversity, compression, and ASR loss. Under standard Lipschitz and strong-convexity assumptions, the expected squared gradient norm satisfies

$$\mathbb{E}[\|\nabla \mathcal{L}(\theta_k)\|^2] \leq \frac{2L(\mathcal{L}(\theta_0) - \mathcal{L}^*)}{\mu k} + \frac{L\sigma_{\text{noise}}^2}{\mu k},$$

yielding an $O(1/k)$ convergence rate with only $O(\log N)$ pruning rounds for an initial dataset of size N .



Validation

1. **Quality over quantity:** Pingala v1 shows that strategically curated, high-entropy data can reach state-of-the-art ASR. Models trained on this subset match or exceed systems trained on much larger, loosely filtered corpora, achieving comparable or lower WER on standard, noisy, and code-mixed benchmarks with orders of magnitude fewer GPU hours.
2. **Multilingual effectiveness:** Entropy-based principles generalize across 200+ languages. The same methodology improves low- and mid-resource languages without degrading high-resource ones. Multilingual training in Pingala v1 enhances rather than degrades performance, preserving strong English results while significantly boosting under-represented languages and code-switching.
3. **Cross-domain robustness:** Pingala V1 maintains stable performance across meetings, earnings calls, broadcast media, read speech, and contact-centre audio, avoiding the large error spikes common in many large-scale ASR models. This indicates that it learns universally valuable acoustic-linguistic patterns rather than overfitting to any single domain.



Impact



Accuracy: Pingala v1 reaches 3.10% WER, matching or surpassing current state-of-the-art ASR.

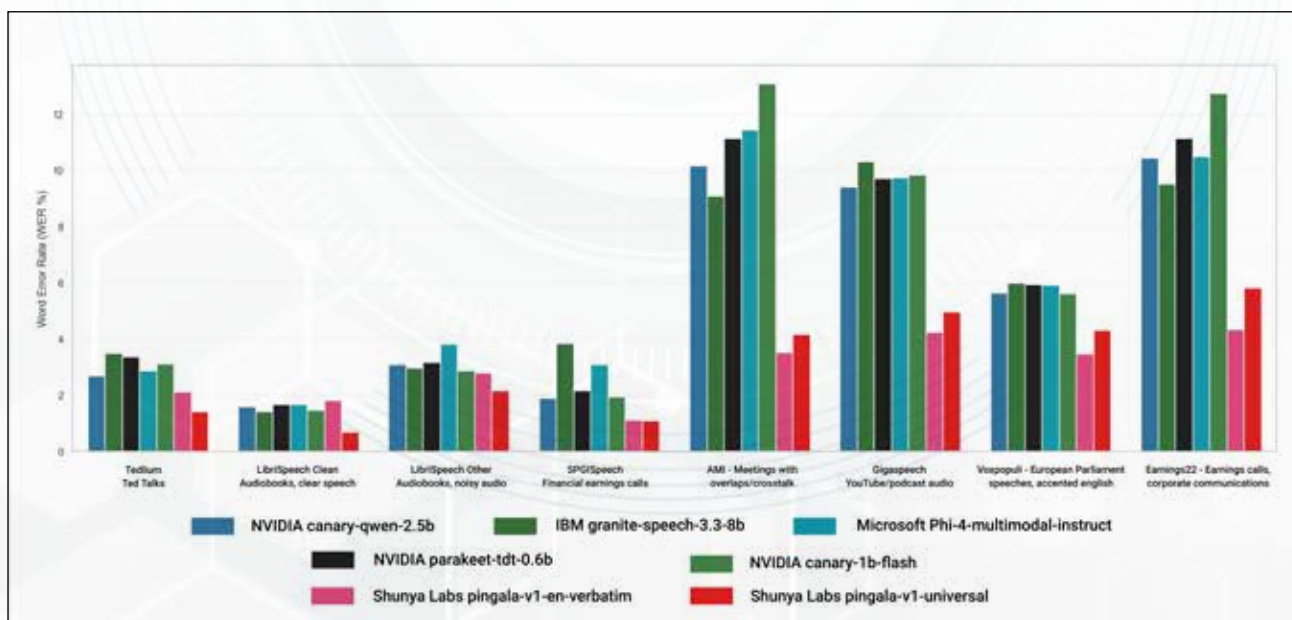


Benchmark standing: It ranks at the top of the OpenASR-style leaderboard, across diverse, real-world test sets.



Latency: Delivers <100 ms end-to-end latency for real-time streaming ASR at high concurrency.

Pingala V1: World's Most Accurate ASR



μPADs: Miniature Paper Based Bio Sensing Devices



Presenter

Dr. Pranjal Chandra, Associate Professor and Ramanujan Fellow Coordinator, School of Biochemical Engineering, Indian Institute of Technology (BHU) Varanasi



Summary

Traditional analyte detection platforms such as screen-printed electrode-based platforms and benchtop analyzers are expensive, bulky, and time-consuming. They often require skilled manpower, laboratory infrastructure, and high operational costs, limiting their accessibility in resource-constrained settings. Moreover, these devices are not eco-friendly and are unsuitable for large-scale, rapid screening applications in healthcare, food safety, and environmental monitoring.

The presented research asset is an industry-ready, laser-directed papertronic electrochemical device (LD-ePAD) designed

for rapid, cost-effective, and sustainable molecular sensing across multiple industries. It represents a significant leap in translating laboratory biosensing technologies into real-world, deployable solutions. Laser engraving technology possesses many advantages, such as high-precision robotic patterning, user-friendly operation, remarkable reproducibility, and minimal waste generation. The LD-ePAD addresses a multi-billion-dollar global need for affordable diagnostics and monitoring tools. By lowering costs and increasing accessibility, it enables adoption in emerging markets, rural regions, and large-scale public health programs. Its integration with cloud analytics can provide recurring business opportunities through subscription-based services and customized industrial solutions.



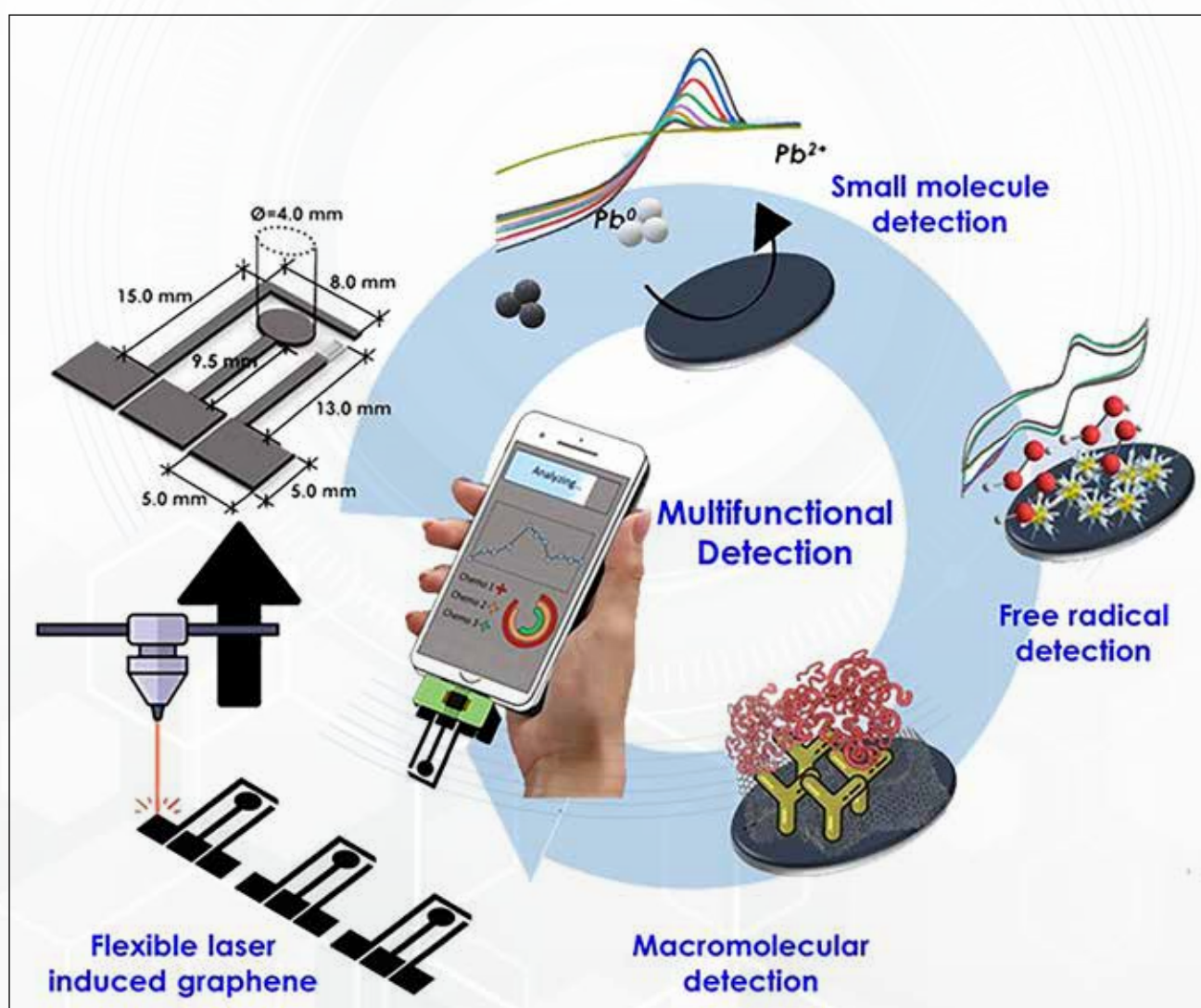
Methodology

The proposed solution utilizes a Laser-Directed Electrochemical Paper Analytical Device (LD-ePAD) that employs normal paper, which is sprayed with a fire-retardant

solution and coated with wax before being laser engraved. Real and graphical illustration of the LD-ePAD along with the SEM image is depicted in Figure 2. This laser-directed electrochemical paper analytical device can do multifunctional analysis, including

small molecule detection to macromolecule detection including free radical detection (Figure 1). This device also demonstrates high sensitivity, accuracy, and stability under real-world conditions.

Figure 1: Schematic showing the development of LD-ePAD and its deployment in multifunctional sensing of small molecule, free radical, and macromolecule, respectively [Reproduced with permission from Mahapatra et al., Chemical Engineering Journal 486, 150232, 2024].





Novelty

This is the first study wherein we have documented the use of controlled one-step laser direct engraving to fabricate a three-electrode setup on a paper substrate that can sense three heterogeneous molecules electrochemically. The published literatures also place a strong emphasis on the sensing of a single type of molecule and lack the potential to detect multifunctional molecules. Additionally, the existing system uses various enzymes to generate analytical signals. Further, the majority of the reported paper-based electrodes primarily deploy a two-electrode system, i.e., the working and counter electrodes. Nevertheless, the lack of a third electrode, i.e., the reference electrode, makes it more difficult to obtain accurate and repeatable results due to the potential drift.

Paper-based diagnostic devices market is about a \$20B opportunity globally, growing at 8% CAGR by the year 2030. The need for miniature paper-based biosensing device for Alkaline Phosphate (ALP) determination is needed across industry including milk pasteurization indication, gall bladder diseases, live diseases, bone cancer, chronic kidney disorders to mention a few. Most of the currently available devices in the market are high-end instruments, with each electrode set costing over \$4 each, and they involve multistep material synthesis and complex processes, besides these chips are not bio degradable. The proposed Papertronic solution uses waste paper discs obtained from the office

punching machine and does not require any instrument for sensor development and qualitative analysis; it can be operated by an untrained person, which is applicable in villages as well as in industries, and it is an environmental friendly, biodegradable system.



Validation

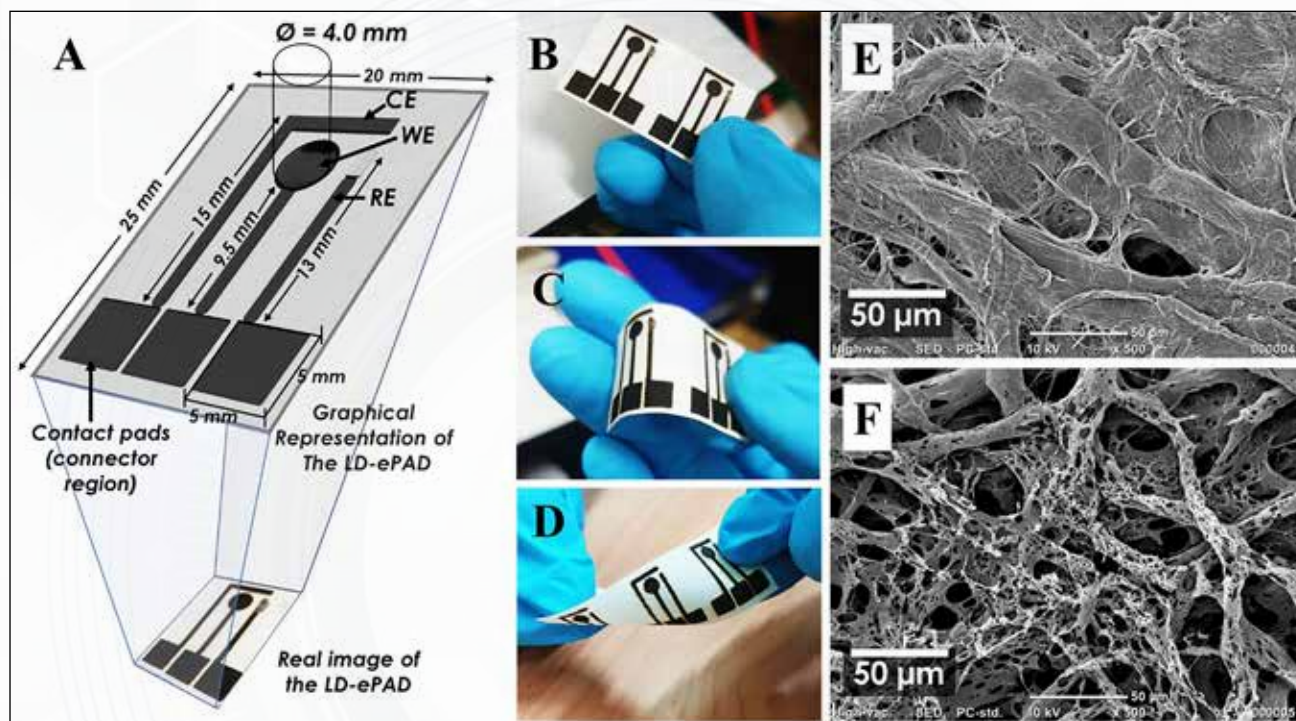
This device has advanced from the prototype to the proof-of-concept stage and has been successfully tested for portability, multifunctional sensing ability, and robustness. Validation studies confirm its capability to replace costly electrodes and provide reliable results in real-world settings. From a field-testing perspective this device has been tested for disease biomarker, contaminant such as heavy metal, , and free radical as preservatives in food item. This device can also be used for environmental monitoring viz rapid on-site detection of pollutants and peroxides in water and soil. This versatility ensures strong alignment with healthcare, food technology, and environmental industries.



Impact

This research device features an interactive user interface, and the ability to deliver results 10x faster than existing diagnostic tools. But more importantly, this device costs only ~25 INR, making it 95% cheaper than commercial screen-printed electrodes

Figure 2: (A) Real and graphical illustration of the LD-ePAD. The CE, WE, and RE are marked appropriately; (B, C, and D) The in-hand depiction of the developed electrode showing miniaturized, flexible, and twistable properties; (E and F) SEM micrographs of the bare paper unmodified surface and the LD-ePAD surfaces, respectively [Reproduced with permission from Mahapatra et al., Chemical Engineering Journal 486, 150232, 2024].



(~3500 INR). Besides this, it has an eco-friendly design, as Papertronic construction makes it environmentally sustainable, lightweight, and disposable. This research demonstrates high sensitivity, accuracy, and stability under real-world conditions along with scalability, as it can be easily mass-manufactured with minimal infrastructure and is adaptable to multiple industries.

This technology democratizes access to affordable diagnostics and monitoring. Its low cost, portability, and sustainable nature ensure that underserved communities gain access to timely healthcare and safer food and environments. By reducing dependence on centralized laboratories and costly devices, the asset contributes to UN

Sustainable Development Goals (SDGs) in health, sustainability, and innovation.



Future Scope

Future scope for this work may include testing this device for other applications particularly in the healthcare segment for point-of-care testing for infectious diseases, biomarkers, and routine diagnostics. It can also apply to food safety for the detection of contaminants, antibiotics, and preservatives in consumables. Equally important is environmental monitoring with rapid on-site detection of pollutants and peroxides in water and soil.

GRAMA

Algorithms for Modern Agriculture with AI & Game Theory



Presenter

Dr. V Priyanka from Indian Institute of Science, Bangalore



Collaborators

This research is in collaboration with Dr Manojkumar P, Dr Rohit S, Dr Geetha Charan and Dr Y Narahari, Honorary Professor at the Department of Computer Science and Automation, Indian Institute of Science Bangalore.



Summary

The GRAMA project (Game Theory, Random Processes, AI, and ML for Agriculture) is focused on how to improve agricultural outcomes for Indian farmers. Fundamentally, this research is directed to address problems such as crop planning and giving advice to farmers on all decisions that the farmers are

called upon to take during the crop life cycle. This constitutes several innovations covering Crop planning system (CROPS), Agri crop recommendation (ACRE), explainability of AI models for agriculture using cooperative game theory (XAI-CGT), Price prediction for agriculture (PREPARE), Protocols for optimal selling of produce for enhanced revenue (PROSPER), procurement mechanisms for inputs and services (PROMISE), India oriented agri-advisories with explainability (KRISHI-X) and AI/ML pipeline for agriculture (AGRI_VAAHAN). The research has been supported by the Indian Institute of Science, NABARD, and BEL.



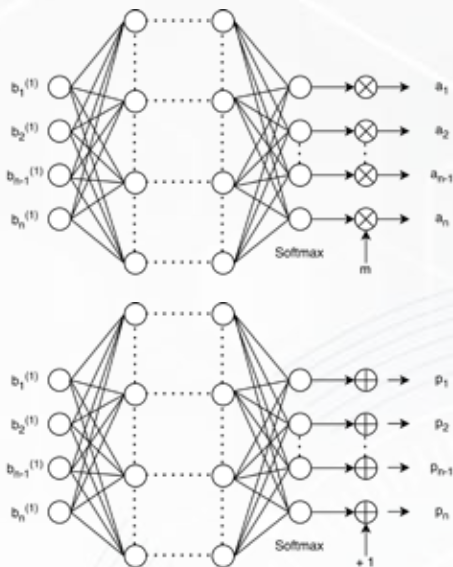
Methodology

The GRAMA project is an overall technology enabled solution for farmers by exploiting key data sources available from Govt of India. For instance, data sources include Digital India Agriculture data from ministry of agriculture & farmers welfare, price data from Agmarknet, Weather data from Indian Meteorology Department, and crop data from package of practices from state agricultural universities and central agricultural



universities. One of these solutions PROMISE (Procurement Market for Agri Inputs and Services) harnesses the potential of farmer producer organizations (FPOs) to achieve cost effective and quality assuring procurement of agricultural inputs such as seeds, fertilizers, and pesticides. Essentially, the inputs required collected through a mobile app are clustered and classified into buckets which are then bulk procured from vendors with a dynamic AI algorithm to minimize cost of procurement but at the same time ensuring quality. PROSPER on the other hand gathers information about farmers which is clustered and classified in to buckets and each bucket is bulk sold to high value buyers again with AI based optimization to dynamically decide the best price and maximizing revenue for farmers. Modules such

as volume discount auctions for PROMISE and PROSPER are based on the design of fair and efficient procurement process which in turn involves AI algorithms for adhering to minimum and maximum supplier limits ensuring practicality. The algorithm is trained to achieve incentive compatibility, individual rationality, social welfare maximization, cost minimization, fairness of allocation, and other business constraints. The crux is the deep learning-based method to learn a volume discount auction that satisfies IC, IR, cost minimization, envy minimization and business rules in a regret minimization sense. Deep Learning volume discount auction that satisfies IC, IR, cost minimization, envy minimization, and business rules in a regret minimization sense is as follows:



$$\text{loss} = \text{cost} + \text{penalty}_{\text{regret}} + \text{penalty}_{\text{envy}} + \text{penalty}_{\text{business}} + \text{LagrangianLoss}$$

$$\text{cost} = \sum_{i=1}^n p_i(b)$$

$$\text{penalty}_{\text{regret}} = \rho_{\text{regret}} \sum_{i=1}^n \tilde{r}_i^2$$

$$\text{penalty}_{\text{envy}} = \rho_{\text{envy}} \sum_{i=1}^n e_i^2$$

$$\text{LagrangianLoss} = \sum_{i=1}^n \lambda_{\text{regret}}^{(i)} \tilde{r}_i + \lambda_{\text{envy}}^{(i)} e_i$$

$$\text{penalty}_{\text{business}} = \begin{cases} 0, & a^{(*)} \geq a_{\min}, \\ \frac{\rho_{\text{business}}}{a^{(*)}}, & a^{(*)} < a_{\min} \end{cases}$$



Novelty

This research uses Game Theory, Random Processes, Artificial Intelligence, and Machine Learning for Agriculture. For instance, PROMISE is a DL based method to learn a volume discount auction that satisfies Incentive Compatibility, Individual Rationality, social welfare maximization, cost minimization, fairness of allocation, envy minimization and business rules in a regret minimization sense, adhering to minimum and maximum supplier limits, ensuring practicality. PROSPER on the other hand is a Protocol for Optimal Selling of Agricultural Produce for Enhanced Revenue.



Impact

GRAMA solutions are unique and novel solutions for Indian agricultural market targeting immense benefits for farmers. As it casts a wider net and gets deployed across the state of Karnataka and over time across India, it holds potential promise to benefit millions of farmers and it should also encourage organizing agri sector, making it more transparent and lucrative business, thus help making India more self-reliant.



Validation

The solution PROMISE is already deployed at rural area FPOs (Anekal and Rajaghatta) across Karnataka along with front end mobile app designed by BEL, backed by auction algorithm from IISc where vendors from New Tharugupet have participated



Future Scope

Further research is required to scale the deployment by training the AI modules with various data sets from across India to improve accuracy and cast a wider net from deployment perspective. This research can also unearth more relevant use cases for the agriculture sector.

Green AI

Low-power Quantized Models or Distilled LLMs Optimized for Compute



Presenter

Mr. Naushath Raja Mohammed, AI Consultant



Collaborators

This research is in collaboration with Chandrasekaran Subramanian (E&Y), Prasana S (Sutherland), Ashok Kumar Arunachalam (PWC) & Adnan Bayoun (AI Consultant - Qatar).



Summary

The Energy-Efficient LLM Training & Inference has the potential to redefine how AI is adopted in India and other resource-constrained regions. By reducing the compute and energy requirements for LLMs, this research directly democratizes access to advanced AI, enabling startups, academia, and rural innovators to build solutions without relying on costly cloud infrastructure or high-end GPUs.



Methodology

Deploying quantized and distilled LLMs on edge devices, affordable GPUs, and even mobile platforms will empower industries such as healthcare, agriculture, BFSI, and education to leverage AI in real-world scenario, ranging from medical diagnostics in rural clinics to personalized learning in regional languages. Beyond accessibility, the project promotes sustainability by lowering AI's carbon footprint, aligning with India's national digital and green goals. It positions India as a global leader in responsible and inclusive AI innovation, ensuring the benefits of LLMs reach the widest possible population.

> Large Language Models (LLMs) have become the foundation of modern AI applications, powering breakthroughs in conversational agents, knowledge automation, and multimodal intelligence. However, training and deploying these models comes at a steep cost, both in terms of compute resources and energy consumption. Traditional LLM pipelines demand access to high-end GPUs and massive infrastructure, creating barriers for organizations and researchers in regions

with limited resources. In a country like India, where digital inclusion is central to innovation, the challenge of high compute cost directly limits the democratization of AI.

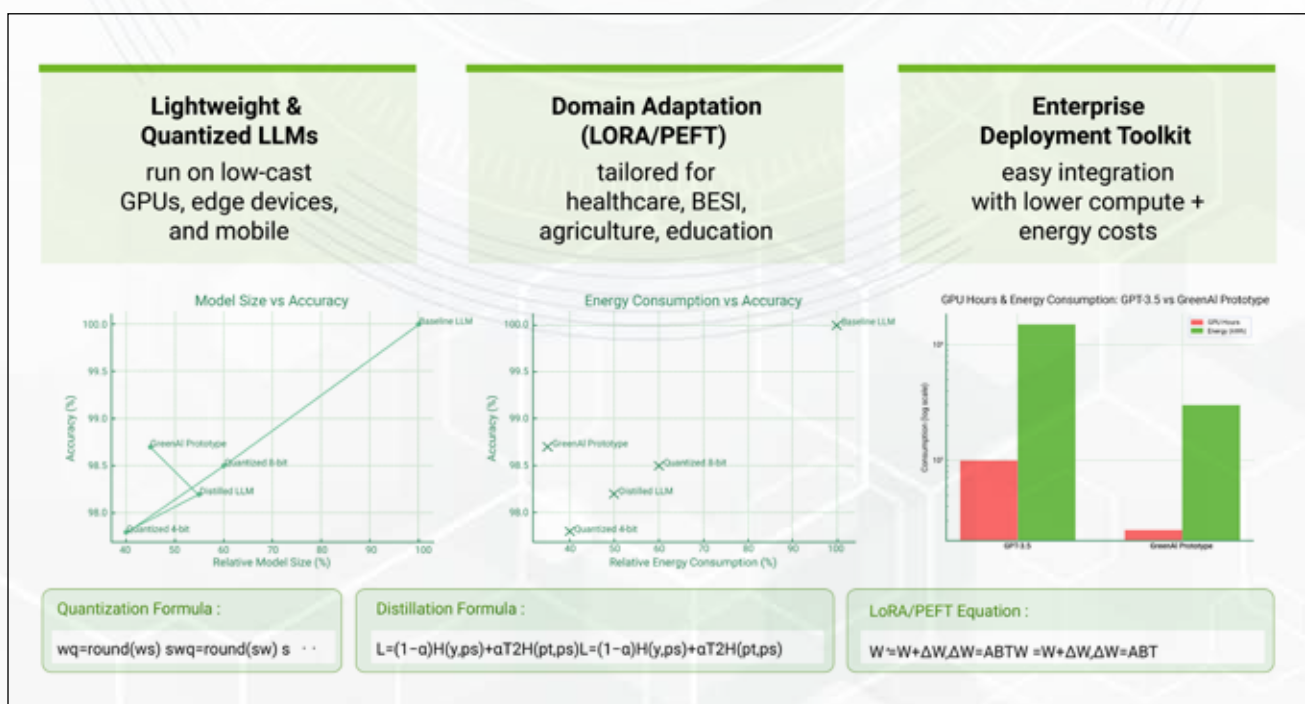
The two core approaches being explored are quantization and distillation. Quantization reduces the precision of model weights and activations, enabling models to run on smaller hardware footprints without significant loss of accuracy. Distillation, on the other hand, transfers knowledge from a large “teacher” model into a smaller “student” model, dramatically reducing compute requirements for inference while maintaining performance on critical tasks.

While global tech leaders often focus on pushing model sizes larger, this initiative reframes the problem by asking: How can we make AI smaller, faster, greener, and more inclusive? The outcomes of this work could position India as a global pioneer in sustainable AI innovation, offering scalable

solutions not just for domestic use but also for emerging markets worldwide.

The research primarily covers:

- Model Optimization Framework** including quantization , INT8 / INT4 and mixed-precision training for compute and memory efficiency & knowledge distillation thus transferring capability from large teacher models → compact student LLMs.
- Low-Rank Adaptation (LoRA / QLoRA)** with efficient fine-tuning with minimal parameter updates.
- Edge & Low-Cost GPU Adaptation** including model runtime optimization with g ONNX Runtime, TensorRT, and PyTorch Mobile, layer fusion & dynamic quantization for deployment on A10, RTX-class GPUs, Jetson, and mobile NPUs & cross-compilation and kernel auto-tuning for heterogeneous hardware targets.



- (d) Domain Adaptation Research including building domain-specific adapters (healthcare, BFSI, agriculture, education) by leveraging parameter-efficient fine-tuning (PEFT) with minimal GPU cycles & training on curated domain datasets + open knowledge graphs to improve factuality.
- (e) Energy-Aware Training Paradigm through carbon tracking metrics integrated into training (power, GPU hours, CO₂ impact), dynamic batch size and learning rate scaling based on energy efficiency curves & collaboration with GreenAI and MLCommons for benchmarking eco-metrics.
- (f) Deployment & Tooling with sustainable LLM Toolkit (S-LLM Kit): compression, quantization, & benchmarking modules, containerized micro-deployment stack for on-prem, edge, or cloud & integration APIs for LangChain, Hugging Face, and ONNX Hub.



Novelty

By combining these techniques with innovations such as low-rank adaptation (LoRA), parameter-efficient fine-tuning (PEFT), and pruning, this research creates lean, high-performance LLMs optimized for India's compute infrastructure. Such models can be trained with fewer resources and deployed on edge devices, affordable GPUs, and even mobile platforms, significantly widening access to AI. This has the potential to transform industries like healthcare,

agriculture, education, and BFSI, where resource-constrained environments often limit technology adoption.



Validation

This Green-AI research asset Energy-Efficient LLMs is lightweight, quantized, and distilled models optimized for low-cost GPUs, edge, and mobile devices, while retaining near-state-of-the-art accuracy while reducing compute and energy consumption by up to 70-80% and is built for real-world scalability in constrained environments. It has domain adaptive intelligence and custom fine-tuning using LoRA / PEFT for key verticals such as healthcare, BFSI, agriculture and education. It enables contextual accuracy and domain fluency without retraining full models and supports multi-lingual and low-resource language adaptation for broader accessibility. This research offers sustainable AI Deployment Toolkit for model compression, inference optimization, and deployment on enterprise stacks with plug-and-play integration with existing cloud or on-prem systems.

Includes energy metrics, performance dashboards, and model governance modules.



Impact

The impact hence is two folds, expanding adoption of AI by making it affordable

and practical while ensuring environmental responsibility, thereby setting a benchmark for sustainable AI worldwide. This research stands out because it solves one of the most pressing challenges in AI such as balancing scale, accessibility, and sustainability. While global tech leaders often focus on pushing model sizes larger, this initiative reframes the problem by asking: How can we make AI smaller, faster, greener, and more inclusive? The outcomes of this work could position India as a global pioneer in sustainable AI innovation, offering scalable solutions not just for domestic use but also for emerging markets worldwide.

India's data-center GPU market valued at USD 794 M (2024); projected CAGR $\approx$ 36% (2025–2033) thus signalling explosive demand however with imposed licensing limits on high-end AI chips (H100/A100), classifying India as a “Tier 2” country we need low power compute to become self reliant. AI publication totals are rising sharply worldwide (AI share of CS publications rose from ~22% in 2013 to ~42% in 2023) building strong research momentum globally and in India. However share of AI scientific publications (2023): China 36.05%, India 12.13%, United States 11.98%. Export controls and GPU scarcity limits India's access to high-end chips (H100/A100); software efficiency (quantization, distillation, PEFT) mitigates dependence on restricted hardware. And hence what we need is lightweight models run on low-cost GPUs or edge devices, cutting CapEx, energy use, and cloud costs which are vital for startups and enterprises.



Future Scope

Further research collaboration & Validation can be explored by partnering with academia and national labs for energy-profiling studies, continuous benchmarking on MLPerf, Green500, and TCO baselines & publishing findings under “Sustainable Compute Initiative (SCI)” framework. Continued research is needed into lightweight model architectures and energy-efficient training techniques. Build a testbed for validating model performance and sustainability metrics. Launch pilot programs with deployment in a controlled environment to gather performance insights. And finally, iteration & optimization to refine based on pilot results for broader rollout and scale & monitor to expand deployment while monitoring performance and sustainability continuously.

Certain use cases include healthcare (accelerating diagnostics with energy-efficient AI while reducing operational costs), finance (enabling real-time analytics and fraud detection without heavy compute demands), manufacturing (optimizing production workflows with low-resource AI solutions), education (delivering personalized learning at scale for emerging markets) & Government & public sector (supporting sustainable AI adoption aligned with national digital transformation goals).

Dhvani

An Automated & Interactive Govt. Form Filling for Aged & Monolingual People



Presenter

Dr. Deepamala N, Professor, RV College of Engineering, Bengaluru, India



Collaborators

This research is in collaboration with Hegde AR, Sujala Reddy RS, Kruthika P, Pragathi BC, Sai Lahari S, Deepamala N, Shobha G. Automated government form filling for aged and monolingual people using interactive tool. Disability and Rehabilitation: Assistive Technology. 2024 Aug 17;19(6):2260-70. <https://doi.org/10.1080/17483107.2023.2277825>



Summary

TDhvani is an automated government form filling for aged and monolingual people using interactive tool. This automated tool will solve the problem of aged, illiterate and physically challenged persons filling forms in post offices and banks. According to the

2011 Census, India has 121 major languages in which 22 are scheduled languages. These are spoken across the country with native speakers spread across states. The Government of India has proposed various schemes for the poor and the needy. These schemes are offered in post offices or in nationalized banks for the citizens of India. Majority of the citizens who avail of these schemes are aged and monolingual people. They find difficulty in recognizing the suitable schemes which will benefit them and also in filling out application forms for these schemes. This is not a problem faced only by monolingual people but also a problem of blind or physically challenged people.

In this research project, an interactive voicebot “Dhvani” has been proposed and developed to help the monolingual or aged people explore all the available schemes based on their needs and also help them fill out forms. The chatbot predicts the scheme, fills the form based on user input, and this can be submitted to the post office or other relevant organization. Currently, the project is developed for the Kannada Language, which is widely spoken in Karnataka, India, and can be further extended to other regional languages.

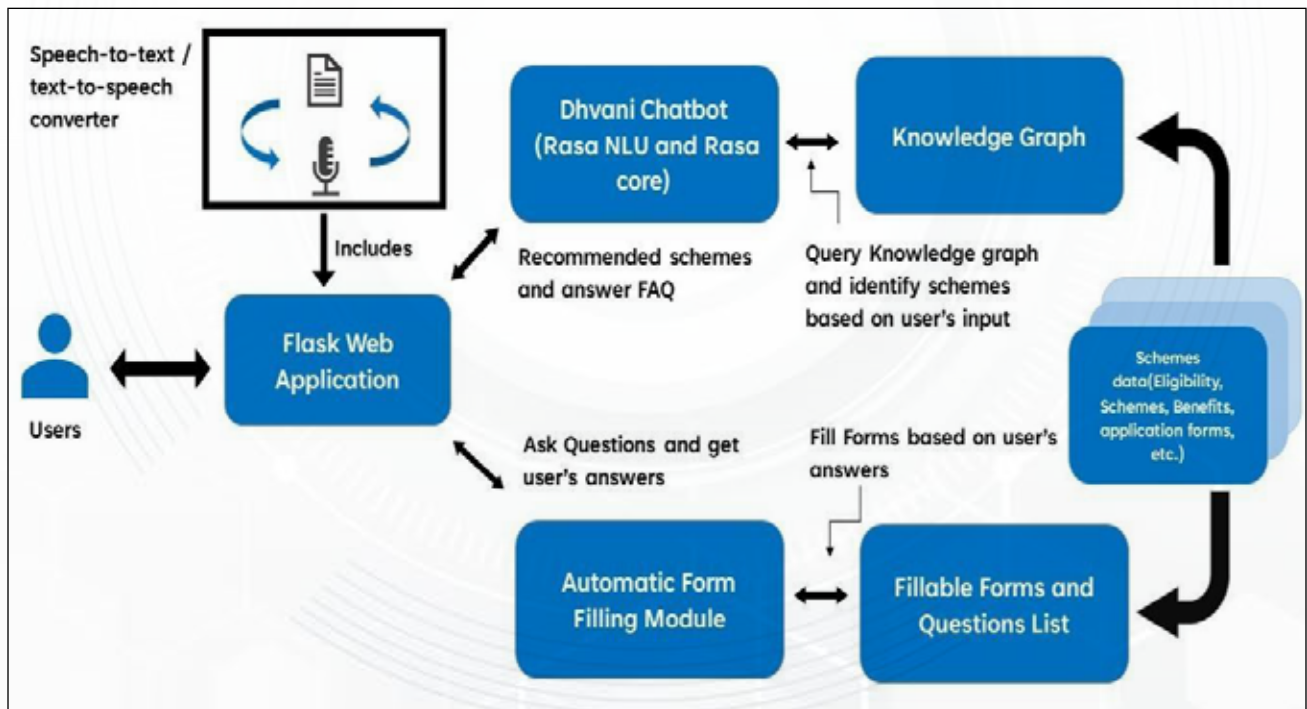


Methodology

The Dhvani voicebot is built using RASA, an open-source framework capable of having layered conversations for building chatbots. The framework consists of RASANLU, Core and other Python libraries to handle intent recognition, entity extraction and conversation flow of chatbot. Chatbot's NLU featurizer is trained from scratch by providing training data. Nearly, 1000 Kannada sentences are manually annotated and used for training purpose.

After the successful prediction of the scheme, the user is given a list of schemes that are suitable, from which the user can choose a scheme.

A database of fillable forms has been created with a questionnaire corresponding for each scheme. For instance live demo of the asset shows that with a query of questionnaire to fill the application for "Sukanya Smruddhi" Scheme, Dhvani enquires the girl child's details and notifies when form is auto-filled.



Novelty

Chatbot's NLU featurizer is trained from scratch by providing training data. Nearly, 1000

Kannada sentences are manually annotated and used for training purpose. A knowledge graph is created to represent semantics by describing entities and their relationships and After the successful prediction of the scheme, the user is given a list of schemes that are

suitable, from which the user can choose a scheme. A database of fillable forms has been created with a questionnaire corresponding for each scheme.



Validation

Performance of the Dhvani voicebot in classifying intents and dialogue management is measured using the F1 score of 0.8 on action prediction & 0.97 on intent classification, while precision score and recall score is also 0.97 each for training set and testing set.

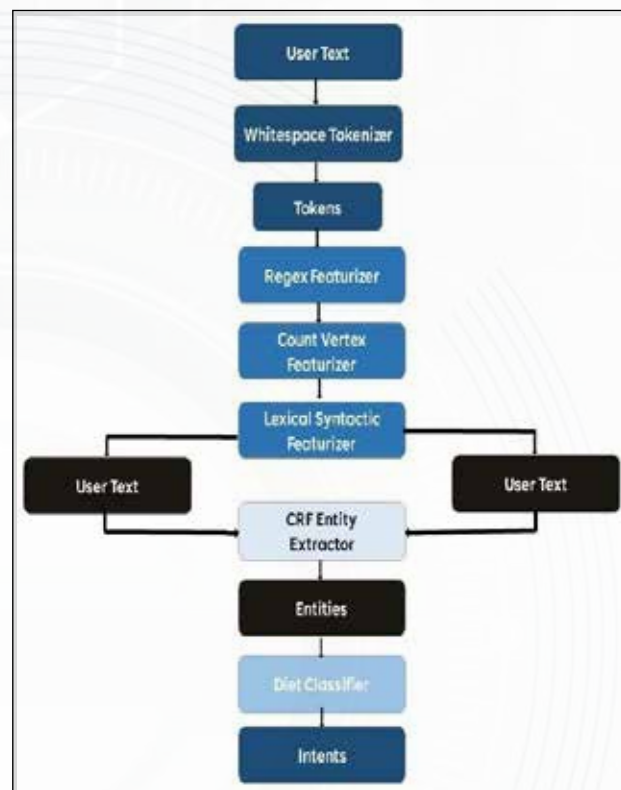
Task	F1 score	Recall score	Precision score
Action Prediction	0.8715	0.8823	0.8834
Intent classification	0.9740	0.9740	0.9752



Impact

This innovation truly reaches out to masses by addressing the challenges faced by monolingual individuals in India, particularly the elderly, people with impairments, and those from marginalized communities and it helps these people connect through voice commands to Govt schemes, which otherwise they would be secluded from. The proposed work though is currently designed for the Kannada language, can be extended

to other languages as well. It helps users in identifying suitable government schemes and fills various Govt Scheme nominations and be able to access it.



Future Scope

Currently, the project is developed for the Kannada Language, which is widely spoken in Karnataka, India, and can be further extended to other regional languages. Future scope also includes use of LLM instead of RASA for chatbot, update more government schemes in knowledge graph database & incorporate more non-Indian languages as well.

Annexure

Speaker Bio

Session - 1



Dr. Sriparna Saha received her M.Tech and Ph.D. in Comp Sc from ISI Kolkata. She is currently an Associate Professor in the Department of Comp Sc & Engg at IIT Patna. She has published over 450 research papers, with an h-index of 52 and more than 13,700 citations (Google Scholar). Her research interests include NLP, ML, DL, Multiobjective optimization & biomedical text mining. She is a Senior Member of IEEE and ACM, and a Fellow of IET (UK) and IETE. She has received several notable awards, including the Google India Women in Engineering Award, NASI Young Scientist Platinum Jubilee Award, SERB Women in Excellence Award, and Fulbright-Nehru Academic and Professional Excellence Fellowship. She also serves as an Associate Editor for leading journals such as IEEE T-CSS, IEEE/ACM TCBB, IEEE Internet Computing, Pattern Recognition, Expert Systems with Applications, and PRL.

sriparna@iitp.ac.in

Session - 2



Dr. Subhasish Sarkar MBBS, MS is posted as Assistant Professor at the Department of General Surgery at College of Medicine and Sagore Dutta Hospital, Kolkata, West Bengal. He had completed his MS from R.G Kar Medical College and Hospital on 2015. He is involved in various academic activities and research. He has more than 12 research papers and conference papers. His lab is dedicated to advancing research in the fields of biomedical engineering, clinical therapeutics, and innovative diagnostics. His primary focus is on developing cutting-edge medical devices and leveraging bioinformatics to address complex health challenges. A central theme of the research is the design and optimization of novel therapeutic strategies for cancer and type 2 diabetes mellitus (T2DM), two of the most prevalent and impactful diseases in global health.

s.sarkar735@gmail.com

Session - 3



Mr. Adarsh Pandey is a doctoral researcher at the Indraprastha Institute of Information Technology (IIIT) Delhi, India. Before joining IIIT Delhi, he worked at the Samsung Research Institute, where he contributed to the Android platform team and developed UI frameworks for large-scale mobile systems. His current research focuses on health monitoring using wearable devices, with a particular emphasis on analyzing physiological signals for clinical insights.

He has been working closely with neurologists to address long-standing challenges in vertigo diagnosis. As part of this effort, he and his team have developed an in-house, low-cost, portable, and intelligent Video Frenzel device that is now used by several neurologists for diagnosing vertigo in real-world clinical settings. His broader research interests span wearable sensing, physiological data analysis, and applied machine learning for healthcare.

adarshp@iiitd.ac.in

Session - 4



Dr. Santhosh Reddy D., Hub Executive Officer of TiHAN, IIT Hyderabad, India's first Technology Innovation Hub for Autonomous Navigation. He earned his Ph.D. in Electrical Engineering from IIT Hyderabad, with research spanning computer vision, deep learning, and autonomous mobility. He has over a decade of experience in ADAS development, AI-driven perception systems, and edge-enabled autonomous technologies. Dr. Reddy has authored more than 25 research publications, contributed to multiple patents, and led innovations such as India's first Autonomous Campus Shuttle and Edge-Enabled Drone Systems. His work continues to advance safe, intelligent, and sustainable autonomous mobility solutions for the future.

tihan.heo@iith.ac.in

Session - 5



Mr. Siddharth Kanungo serves as Technical Advisor to the CEO's office at iHub HCI Foundation, IIT Mandi, where he contributes to strategic technology initiatives and innovation programs. As founder of Primerlabs, he has navigated seven years of strategic transformation aligned with the evolving technology landscape; transitioning from conversational learning interfaces to HR solutions for employee onboarding and training, and currently focusing on enabling companies to develop prototypes for AI transformation. His diverse client portfolio spanning blockchain ventures, Silicon Valley technology companies, and a leading Palo Alto startup studio has equipped him with distinctive insights into how organizations across scales successfully adopt emerging technologies while sustaining robust learning and development capabilities.

sahil@ihubiitmandi.in

Session - 6



Dr. Rajat K. De is a Professor (HAG) of the Indian Statistical Institute, Kolkata, India. He obtained his Ph.D. degree from the same institute in the year 2000. He was a Distinguished Postdoctoral Fellow of the Whitaker Biomedical Engineering Institute, the Johns Hopkins University, USA, during 2002-2003. He visited the Department of Medicine, the University of California, San Diego, USA in the years 2017 and 2018, with a Fulbright-Nehru Academic and Professional Excellence Fellowship. He has been working in the area of machine learning, bioinformatics, computational biology, and systems biology. Professor De has more than 100 research papers published in international journals, conference proceedings and in edited books, and co-edited three books.

rajat@isical.ac.in

Session - 7



Mr. Balaji Uppili, CBO - India & UK, Neurealm, has over 32 years of experience in the IT industry, across application value management, infrastructure management, BPO and Strategic Operations handling, P&L, Delivery, Operations and customer relations across multiple verticals which includes Information Services, Retail & Hospitality, Media & Publishing, Manufacturing & Logistics, and Consumer goods. Balaji has played various roles across geographies viz. USA, Continental Europe, and Asia Pacific, through his previous stints at Patni Computers, L&T Infotech, Cognizant Technology Solutions, and Virtusa Software Solution Limited. He has also been actively involved in shaping AI strategy and driving AI-led transformation initiatives for customers across these industries.

balaji.uppili@neurealm.com

Session - 8



Dr. Karthik Vaidhyanathan is an Assistant Professor at the Software Engineering Research Center, IIIT-Hyderabad, where he is also associated with the leadership team of the smart city living lab. He obtained his Ph.D. from the Gran Sasso Science Institute, Italy and did his postdoc at the University of L'Aquila, Italy. His main research interests lie in the intersection of software architecture and AI with specific focus on building sustainable software systems. His research focuses on how AI techniques can be leveraged to better architect self-adaptive systems and further how to better define architecting practices for developing AI-enabled systems. He is also an editorial board member (SE Radio) of IEEE Software. Karthik also has more than 5 years of industrial experience as an employee and as a consultant in building and deploying ML products/services.

karthik.vaidhyanathan@iiit.ac.in

Session - 9



Dr. Pankaj Dayama is a Senior Technical Staff Member & Senior Manager at IBM India Research Lab, where he leads the AI for Industrial Solutions department. His current focus is on developing advanced techniques for time series analysis, specifically tailored for industrial applications and IT operations. He is passionate about technology-driven problem solving with a strong emphasis on validating innovations and infusing them into real-world products. Pankaj has authored over 30 research papers in peer-reviewed international conferences and journals, and holds more than 80 filed patents with the USPTO. Prior to joining IBM, he worked as a Research Scientist at General Motors R&D, India. He earned his M.S. and Ph.D. in Computer Science from the Indian Institute of Science (IISc), Bangalore.

pankajdayama@in.ibm.com

Session - 10



Dr. Sachit Rao obtained his Master's and Ph.D. degrees in Mechanical Engineering from the Ohio State University. His specialization is in the area of Control Systems and Theory with an emphasis on Sliding Mode Control.

He has over 3 years of post-doctoral experience working in the areas of control of electric machines, swarms of dynamic agents, and guidance laws for tactical interceptors. He also has 3 years of industrial experience working for an Indian robotics company. Prior to joining IIITB, he was a faculty in the Aerospace Engineering department of Jain University for a year. His interests lie in the design and implementation of controllers for robotic manipulators and unmanned systems. He works with industry in developing Autonomous Mobile Robots.

Sachit@iiitb.ac.in

Session - 11



Mr. Manikandan Padmanaban obtained his Bachelors in E&E from Anna Univ Chennai and M. Tech in power and Energy systems from IIT Kharagpur. He is a Senior Research Engineer and Master Inventor at IBM Research India. He leads development of solutions and research prototypes that address real-world challenges in climate and sustainability, with a particular focus on geospatial AI. He is leading development of a technology stack aimed at tackling common enterprise challenges on their path to achieving net-zero carbon emissions. This involves a comprehensive approach to decarbonization across various facets of their processes and operations. He is an IEEE Senior Member with a strong professional record, holding over 90 USPTO patent filings and authoring more than 30 publications in peer-reviewed conferences and journals.

manipadm@in.ibm.com

Session - 12



Mr. Sourav Banerjee is the Founder and CTO of United We Care and Shunya Labs, where he currently leads teams building deep-tech voice AI inference layers, medical ASR, and domain-specific language models for healthcare and enterprise. He is currently pursuing a PhD in Human-Computer Interaction at IIT Kharagpur. His research focuses on safe and reliable AI systems; he authored “LLMs Will Always Hallucinate, and We Need to Live With This,” a widely cited paper on structural hallucinations in large language models published in Nature. Sourav holds multiple patents and has authored several peer-reviewed papers published in international journals and conference proceedings.

Sb@shunyalabs.ai

Session - 13



Prof. Dr. Pranjal Chandra is currently Head and an Associate Professor at the School of Biochemical Engineering at the Indian Institute of Technology (BHU) Varanasi, India. He earned his Ph.D. from Pusan National University, South Korea, and did postdoctoral training at Technion-Israel Institute of Technology, Israel. With over 15 years of professional experience, he focuses on interdisciplinary research in biotechnology, nanobiosensors, and biochemical diagnostics. His group has designed several commercially viable biosensing prototypes that can be operated for onsite analysis for diverse applications. He has two international patents (South Korea), four national patents granted and has authored more than 200 journal articles/invited chapters. He has published more than 20 books on various aspects of biosensors/diagnostics/material engineering from IET London, Springer Nature, and CRC Press USA.

pranjal.bce@iitbhu.ac.in

Session - 14



Ms. Priyanka V is a Research Associate at the Game Theory Laboratory, Indian Institute of Science, Bangalore, and holds a Ph.D. in Agricultural Economics from UAS Bangalore. Her research focuses on production economics, econometrics, optimization of cropping patterns, and AI/ML-based agricultural decision-making, strengthened by her training in game theory, stochastic models, and computational optimization. She has authored over 12 research papers, along with multiple book chapters and conference publications in agricultural economics analysis. She has delivered guest lectures at Christ university and IIHR Bangalore. Priyanka has been honoured with the Best Ph.D. Thesis Award, ICSSR Doctoral Fellowship, ICAR PG Fellowship, and Best Oral Presentation Award at the Post Graduate Science Week, reflecting her consistent academic excellence and research contributions.

priyankav@iisc.ac.in

Session - 15



Mr. Naushath Raja Mohammed is a seasoned AI and ML leader with over 17 years of experience in developing automation platforms and delivering 100+ customer-focused products. As head of the AI Center of Excellence, he drives enterprise-wide AI adoption and holds advanced certifications from the Indian Defense Institute of Advanced Technology and IIT Madras, along with ongoing NLP research with Liverpool John Moores University. Recognized among Tradeflock's "10 Best Leaders from AI in India 2023," he also serves on the NASSCOM AI Advisory Board and the GenAI Foundry Panel. His recent work on Green AI—focused on sustainable, energy-efficient models—was ranked among India's Top 16 research projects for 2025, highlighting his commitment to responsible AI innovation

Naushath6001@gmail.com

Session - 16



Dr. Deepamala N., Professor, RV College of Engineering has PhD in NLP under VTU, Belagavi in 2016. Her major fields of study include NLP, Computer Networks.

For over five years, she has worked in Software Engineering roles in D-Link, Nortel Networks, and Radware. Later, she joined the RVCE Bengaluru, India, has worked for 16 years in teaching. She has two patents filed and has multiple publications which are Scopus and WOS indexed. She has carried out multiple consultancy and research projects for IBM, Drishti IIT Indore, DRDO, SCII etc. She has been invited speaker in various events including AI conference on Artificial Intelligence & Machine Learning, organized by Mindspace Conferences, Copenhagen, Denmark.

gshobha@rvu.edu.in

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